A pole-free torus one-point Virasoro-block recursion at $c = \frac{1}{2}$ via exact algebraic confluence

Abstract

We construct a coefficientwise finite confluence formula for the eta-reduced torus one-point Virasoro block at the Ising central charge. Degenerate weights and their collision classes are classified exactly. At every grade, generic Kac recursion terms are grouped by their common Ising weight, and each fixed-central-charge Laurent coefficient is an exact collision moment: the constant term of a finite sum of Laurent germs. A valuation bound proves that only finitely many moments occur, an a priori recursive majorant gives a finite jet budget, and a grade-decreasing shifted recursion computes them. The Laurent parameter is an exact algebraic device, not a numerical regularization. Rational continuation and partial-fraction uniqueness prove coefficientwise equality with the inverse-Shapovalov definition to all orders in q. Local Smith and crossing-form results identify the representation-theoretic meaning of the moments. The earliest collision occurs at level three, in the class of the energy weight $\Delta = \frac{1}{2}$, and its embedding diamond produces the first double pole at grade seven. Both its residue and its complete principal part are given by finite fixed-c formulas. The spin-module collision at grade four and its grade-ten diamond provide a second, larger test. As a separate quotient consistency check, the corresponding BGG resolutions reproduce the three Ising characters.

Keywords: Virasoro algebra; conformal blocks; torus one-point block; Zamolodchikov recursion; minimal models; Ising model.

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1 Introduction

Residue recursions for Virasoro blocks originate with Zamolodchikov's recursive representations of the sphere four-point block [1, 2]; see [6] for a modern general treatment. For generic central charge, the eta-reduced torus one-point Virasoro block obeys the residue recursion conjectured by Poghossian [5] and derived in [8, Section 3, equations (7)–(9)]. At rational values of the Coulomb-gas parameter, distinct Kac labels have the same degenerate weight and the individual generic-recursion summands develop parameter poles. Javerzat, Santachiara, and Foda analyzed the cancellation of these non-physical poles in examples and in the identity-insertion character problem [9, Sections 4–6, especially equations (60)–(61)]. Stocco subsequently constructed central-charge-pole-free torus expressions through fourth order and proposed an all-order Laurent-expansion procedure, whose cancellation and algebraic form were left conjectural [10, Sections II.2–II.3 and Appendix A]. Ribault identifies a recursion at rational β^2 as an open problem and singles out torus one-point Virasoro blocks as its simplest incarnation [12, Section 4.3.3]. Higher-order internal-weight poles and rational limits of sphere four-point blocks were also analyzed by Ribault [11, Section 3]; those results provide important precedent for the confluence mechanism, although they do not give a finite torus one-point recursion.

This paper solves a precise version of that problem at $c = 1/2$, for the meromorphic Verma-module block, by an exact algebraic confluence. "Pole-free" here means free of the divergent, non-physical parameter poles in the final recursion coefficients; the genuine Laurent poles in the internal weight Δ remain. The construction uses the formal germ $b = b_0 + t$ to combine generic summands before taking a constant term. It therefore produces an exact formula at fixed c, but is not an intrinsic recursion that avoids an off-critical deformation altogether. No numerical ε-limit is taken. At every grade we partition the generic Kac summands into their exact Ising collision classes. Each class has a regular sum, and its fixed-c coefficients are finite constant terms of finite Laurent sums. The shifted recursion decreases the remaining grade, while a recursive valuation majorant supplies an a priori finite jet budget.

In the notation of Sections 4–8, the main result reads: for every grade $N \geq 1$,

$$H_N\big(\tfrac{1}{2}, \Delta, d\big) = \sum_{n\geq 0} \sum_{k=1}^{M_{n,N}} \frac{A_{n,k,N}(d)}{(\Delta - \lambda_n)^k}, \qquad A_{n,k,N}(d) = \mathrm{CT}_t \sum_{a\in\mathcal{C}_{n,N}} R_a(t)\, S_{a,N-\ell_a}(t)\, \delta_a(t)^{k-1},$$

where H_N is the N-th eta-reduced block coefficient, λ_n runs over the Ising degenerate weights, $\mathcal{C}_{n,N}$ is the finite class of Kac labels (r,s) with $rs \leq N$ whose weight collides at λ_n, $t = b - b_0$ is the exact Coulomb-gas germ, R_a and $\delta_a = h_a - \lambda_n$ are the generic residue coefficient and the weight gap of the label a, $S_{a,M}$ is the shifted Laurent block of Definition 8.1, CT_t extracts the constant term of a Laurent germ, and $M_{n,N}$ is the finite structural pole bound of Definition 8.3. Every ingredient is a finite jet of an explicit rational function of b. Theorem 8.6 proves this identity: the constant-term moments on the right are exactly the partial-fraction coefficients of the inverse-Shapovalov block, at every power of q.

The local calculations exhibit how the algebraic confluence records Virasoro embedding geometry. The earliest collision is the transverse energy-module collision at level three; its two branches first meet at level seven and create the first double pole. The fixed-c formulas are Theorems 5.8 and 6.7. The spin-module collision at level four and its larger level-ten diamond are treated in Theorems 5.11 and 7.4. Finally, Theorem 10.3 distinguishes the Verma identity specialization from the irreducible quotient and records the three standard Ising characters as a separate BGG consistency check.

2 Blocks and the reduced series

Definition 2.1 (Virasoro Verma module)**.** The Virasoro algebra has generators L_n, $n \in \mathbb{Z}$, and central element c, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

The Verma module $M(c,\Delta)$ is generated by a vector $|\Delta\rangle$ such that

$$L_0|\Delta\rangle = \Delta|\Delta\rangle, \qquad L_n|\Delta\rangle = 0 \quad (n > 0).$$

For a partition $Y = (y_1 \geq \cdots \geq y_\ell)$, set $L_{-Y} = L_{-y_1} \cdots L_{-y_\ell}$, $|Y| = \sum_i y_i$, and $\ell(Y) = \ell$.

Definition 2.2 (Shapovalov and torus matrices)**.** At level N, let

$$G_N(c,\Delta)_{Y,Y'} = \langle\Delta|L_Y L_{-Y'}|\Delta\rangle, \qquad |Y| = |Y'| = N,$$

where $L_Y = L_{y_\ell} \cdots L_{y_1}$. For a primary field $V_d(z)$ normalized by $\langle\Delta|V_d(z)|\Delta\rangle = z^{-d}$, let

$$\rho_N(c,\Delta,d)_{Y,Y'} = \langle\Delta|L_Y V_d(1) L_{-Y'}|\Delta\rangle.$$

Whenever G_N is invertible, define

$$F_N(c,\Delta,d) = \sum_{|Y|=|Y'|=N} \big(G_N(c,\Delta)^{-1}\big)_{Y,Y'} \rho_N(c,\Delta,d)_{Y,Y'}.$$

Definition 2.3 (Eta-reduced block)**.** Write

$$\mathcal{F}_{\Delta}^{(c)}(d \mid q) = q^{\Delta - c/24} \sum_{N \geq 0} F_N(c, \Delta, d) q^N$$

and define

$$H_{\Delta}^{(c)}(d \mid q) = \prod_{m \geq 1} (1 - q^m) \sum_{N \geq 0} F_N(c, \Delta, d) q^N = \sum_{N \geq 0} H_N(c, \Delta, d) q^N.$$

Lemma 2.4 (Polynomial matrix entries)**.** *For fixed N, every entry of $G_N(c, \Delta)$ belongs to $\mathbb{Q}[c, \Delta]$, and every entry of $\rho_N(c, \Delta, d)$ belongs to $\mathbb{Q}[c, \Delta, d]$.*

Proof. Repeated use of the Virasoro commutator moves every positive mode in L_Y to the right. Positive modes annihilate $|\Delta\rangle$, while each L_0 and the central element act by Δ and c, respectively. This gives the first assertion.

For the second assertion use, in addition,

$$[L_n, V_d(z)] = z^n \big(z\partial_z + (n+1)d\big) V_d(z).$$

Move positive modes through $V_d(z)$, then move the remaining negative modes through $V_d(z)$ toward the bra. The process terminates because each commutation removes one mode from one side of the insertion. The remaining derivatives act on z^{-d}; after setting $z = 1$, their values are polynomials in d. The other commutators contribute only polynomials in c and Δ. $\square$

Remark 2.5 (Terminology and notation)**.** The words "level" and "grade" are used interchangeably for the L_0-offset N. By Lemma 2.4 and Cramer's rule, each F_N and H_N is a rational function of (c, Δ) and a polynomial in d; note $H_0 = 1$. For a rational function f of Δ and a point λ, we write $\mathrm{PP}_{\Delta=\lambda} f$ for the principal part of f at λ, that is, the sum of the negative-power terms of its Laurent expansion, and $\mathrm{Res}_{\Delta=\lambda} f$ for the coefficient of $(\Delta - \lambda)^{-1}$.

Remark 2.6 (Symbol families)**.** For reference, the following letters serve more than one family, always distinguished by their subscripts and context. The letter e_i denotes local Smith exponents (Section 5); e_j, used only inside the proofs of Lemma 4.1 and Proposition 6.4, denotes the coefficients of the Euler product $\prod_{m \geq 1}(1 - q^m)$; $e = (u, v)$ is a Kac label in Section 8; and $\varepsilon_a = \mathrm{ord}_t\, \delta_a$ is the collision order of Definition 8.3. The calligraphic $\mathcal{A}_{r,s}, \mathcal{P}_{r,s}, \mathcal{R}_{r,s}$ are the generic residue data of Definition 6.1, whereas $A_{n,k,N}$ are the fixed-c partial-fraction coefficients of Theorem 4.2 and $\widetilde{A}_{n,k,N}$ the collision moments of Definition 8.3; the two are identified in Theorem 8.6. ρ_N is the torus matrix of Definition 2.2, while ρ_e in Lemma 8.2 is a valuation bound; similarly $p(N)$ counts partitions, while $p_{a,e}$ in Lemma 8.4 is a precision offset. Finally, b is the Coulomb-gas parameter, b_χ the crossing scalar of Lemma 5.6, and β a crossing form; Stocco's $\beta = b^2$ occurs only in Remark 9.1.

3 Ising resonance classes

Definition 3.1 (Ising Kac weights)**.** For arbitrary central charge, write $\Delta_{r,s}(c)$ for the standard Kac degenerate weight. In the Coulomb-gas parametrization

$$c = 1 + 6(b + b^{-1})^2,$$

it is

$$\Delta_{r,s}(c(b)) = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{4}.$$

For $r, s \in \mathbb{Z}_{>0}$, define

$$\Delta_{r,s} = \frac{(4r - 3s)^2 - 1}{48}.$$

Thus $\Delta_{r,s} = \Delta_{r,s}(1/2)$. Set

$$\lambda_n = \frac{n^2 - 1}{48} \qquad (n \in \mathbb{Z}_{\geq 0}).$$

Proposition 3.2 (Exact collision classification)**.** *For positive pairs (r, s) and (r', s'), $\Delta_{r,s} = \Delta_{r',s'}$ if and only if one of the following holds for some $\ell \in \mathbb{Z}$:*

$$(r', s') = (r + 3\ell, s + 4\ell),$$

or

$$(r', s') = (3\ell - r, 4\ell - s).$$

Proof. The equality of the two weights is equivalent to

$$4r' - 3s' = \pm(4r - 3s).$$

If the sign is $+$, then $4(r' - r) = 3(s' - s)$. Since $\gcd(3, 4) = 1$, there is an $\ell \in \mathbb{Z}$ with $(r' - r, s' - s) = (3\ell, 4\ell)$. If the sign is $-$, then $4(r' + r) = 3(s' + s)$, and the same coprimality argument gives $(r' + r, s' + s) = (3\ell, 4\ell)$. The converse follows by direct substitution. □

Corollary 3.3 (Distinct weights and local finiteness)**.** *The set of distinct positive-label degenerate weights at $c = 1/2$ is*

$$\{\lambda_n : n \in \mathbb{Z}_{\geq 0}\}.$$

Every collision class contains infinitely many positive pairs. For each fixed level N, only finitely many classes contain a pair with $rs \leq N$.

Proof. For every positive pair, $\Delta_{r,s} = \lambda_{|4r-3s|}$. Conversely, given $n \geq 0$, choose a sufficiently large positive $r \equiv n \pmod 3$; then $s = (4r - n)/3$ is a positive integer and $4r - 3s = n$. Starting with any positive pair in a class, the pairs $(r + 3\ell, s + 4\ell)$, $\ell \geq 0$, are positive and remain in the same class. Finally, $rs \leq N$ implies $r \leq N$ and $s \leq N$, so only finitely many pairs, and hence classes, occur. □

4 Coefficientwise partial fractions

Lemma 4.1 (Large internal weight)**.** *For fixed c, d, N,*

$$\lim_{\Delta \to \infty} F_N(c, \Delta, d) = p(N),$$

where $p(N)$ is the number of partitions of N. Consequently,

$$\lim_{\Delta \to \infty} H_N(c, \Delta, d) = \delta_{N,0}.$$

Proof. For a partition Y, put $a_Y = \ell(Y)$, and rescale the PBW vector $L_{-Y}|\Delta\rangle$ by $\Delta^{-a_Y/2}$. Commuting positive modes to the right shows that

$$\deg_\Delta G_N(Y, Y') \leq \min(a_Y, a_{Y'}).$$

If $a_Y = a_{Y'}$ and $Y \neq Y'$, the degree is at most $a_Y - 1$. For $Y = Y'$, the leading term is

$$\left(\prod_{j \geq 1} m_j(Y)!(2j)^{m_j(Y)} \right) \Delta^{a_Y},$$

where $m_j(Y)$ is the multiplicity of j in Y. Indeed, in a fully commuted matrix element every factor of Δ arises from an occurrence of L_0, that is, from a contraction of a positive mode of L_Y

against a negative mode of $L_{-Y'}$ of the same index through the $2jL_0$ term of $[L_j, L_{-j}]$. Each contraction consumes one mode on each side, so a single term carries at most $\min(a_Y, a_{Y'})$ factors of Δ; this proves the degree bound. If $a_Y = a_{Y'}$ and the degree is attained, every mode of Y is contracted against a mode of Y' of equal index, which forces $Y = Y'$; hence off-diagonal entries of equal length have degree at most $a_Y - 1$. For $Y = Y'$, the top-degree terms are the perfect matchings of equal indices: the $m_j(Y)!$ matchings of the modes of index j each contribute $(2j\Delta)^{m_j(Y)}$ to leading order, giving the displayed leading term.

Apply the same counting with one V_d insertion. The number of Δ-factors of a term is again the number of positive–negative contractions producing L_0. A term that uses no Ward commutator $[L_n, V_d(z)]$ reproduces the corresponding Shapovalov term exactly, because $\langle\Delta|V_d(1)|\Delta\rangle = 1$; a term that uses at least one Ward commutator has consumed a mode without producing L_0, so its degree is strictly below $\min(a_Y, a_{Y'})$. It follows that, after the above rescaling, both G_N and ρ_N converge to the same invertible diagonal matrix D_N. Therefore

$$\lim_{\Delta\to\infty} F_N(c, \Delta, d) = \operatorname{tr}(D_N^{-1} D_N) = p(N).$$

If e_j are the coefficients of $\prod_{m\geq 1}(1 - q^m)$, then

$$H_N = \sum_{j=0}^{N} e_j F_{N-j}.$$

The generating-function identity

$$\left(\sum_{j\geq 0} e_j q^j\right)\left(\sum_{k\geq 0} p(k) q^k\right) = 1$$

gives the second limit. $\square$

Theorem 4.2 (Finite coefficientwise pole expansion)**.** *At $c = 1/2$, $H_0 = 1$, and for every $N \geq 1$ there are unique coefficients $A_{n,k,N}(d) \in \mathbb{Q}[d]$, only finitely many of them nonzero, such that*

$$H_N\left(\frac{1}{2}, \Delta, d\right) = \sum_{\substack{n\geq 0\\ \exists\, r,s>0:\ rs\leq N,\ |4r-3s|=n}} \sum_{k\geq 1} \frac{A_{n,k,N}(d)}{(\Delta - \lambda_n)^k}.$$

The largest k with $A_{n,k,N} \neq 0$ is the multiplicity of the pole of H_N at λ_n. In particular, setting

$$A_{n,k}(d; q) = \sum_{N\geq 1} A_{n,k,N}(d) q^N$$

gives a coefficientwise locally finite expansion

$$H_\Delta^{(1/2)}(d \mid q) = 1 + \sum_{n\geq 0}\sum_{k\geq 1} \frac{A_{n,k}(d; q)}{(\Delta - \lambda_n)^k}.$$

Proof. By Lemma 2.4, $F_N = \operatorname{tr}(G_N^{-1}\rho_N^{\mathsf{T}})$ is rational in Δ, and its poles are contained in the zero locus of $\det G_N$. The Kac determinant formula [3, 4], in the form of [7, Theorem 4.2 and equations (4.16)–(4.21)], gives, up to a nonzero factor independent of Δ,

$$\det G_N(c, \Delta) = C_N \prod_{\substack{r,s>0\\ rs\leq N}} \left(\Delta - \Delta_{r,s}(c)\right)^{p(N-rs)}.$$

At $c = 1/2$, Proposition 3.2 identifies the distinct zeros with the stated finite set of λ_n's. Since H_N is a finite integer linear combination of $F_0, \dots, F_N$, it has the same pole containment. Lemma 4.1 shows that H_N tends to zero at infinity for $N \geq 1$, so its partial-fraction expansion has no polynomial part. Such an expansion is unique, because a proper rational function is the sum of its principal parts at its finite poles. All matrix entries have rational coefficients at $c = 1/2$, and their dependence on d is polynomial; hence the partial-fraction coefficients belong to $\mathbb{Q}[d]$. Local finiteness follows from Corollary 3.3. □

5 Local Smith exponents, transverse collisions, and the first residues

Lemma 5.1 (Pole order from local invariant factors)**.** *Let $R = \mathbb{C}[[x]]$, and let $G(x) \in M_m(R)$ have nonzero determinant. Suppose the Smith exponents of G are $0 \leq e_1 \leq \cdots \leq e_m$. Then*

$$\operatorname{ord}_x \det G = \sum_{i=1}^{m} e_i, \qquad \dim \ker G(0) = \#\{i : e_i > 0\},$$

and the pole order of $G(x)^{-1}$ is e_m. In particular, if

$$\operatorname{ord}_x \det G = \dim \ker G(0) = k,$$

then every positive Smith exponent equals 1, and G^{-1} has at most a simple pole.

Proof. Since R is a discrete valuation ring, there are $U, V \in \mathrm{GL}_m(R)$ such that

$$UGV = \operatorname{diag}(x^{e_1}, \dots, x^{e_m}).$$

Taking determinants gives the first identity. Reducing modulo x gives the second. Inverting the displayed Smith form gives pole order e_m. If the sum of the k positive integers e_i is k, then each of them is 1. □

Proposition 5.2 (Transverse inverse residue)**.** *Let V be finite dimensional, let $G(x) \colon V \to V^*$ be a symmetric matrix over $\mathbb{C}[[x]]$ with nonzero determinant, and put $K = \ker G(0)$. Let $\iota \colon K \hookrightarrow V$ be the inclusion and define the crossing form*

$$\beta(u, v) = u^{\mathsf{T}} G'(0) v, \qquad u, v \in K.$$

If β is nondegenerate, then $G(x)^{-1}$ has only a simple pole and

$$\operatorname*{Res}_{x=0} G(x)^{-1} = \iota\, \beta^{-1} \iota^*.$$

Moreover, β is nondegenerate if and only if

$$\operatorname{ord}_x \det G = \dim K.$$

Proof. Choose a complement C of K. Since K is the radical of the symmetric form $G(0)$, its restriction A to C is nondegenerate. In the decomposition $V = K \oplus C$, one therefore has

$$G(x) = \begin{pmatrix} xB + O(x^2) & xD + O(x^2) \\ xD^{\mathsf{T}} + O(x^2) & A + O(x) \end{pmatrix},$$

where B is the matrix of β. The Schur complement of the lower-right block is

$$S(x) = xB + O(x^2).$$

If B is invertible, block inversion gives

$$G(x)^{-1} = \frac{1}{x}\begin{pmatrix} B^{-1} & 0 \\ 0 & 0 \end{pmatrix} + O(1),$$

which is the coordinate expression of the asserted residue. Also

$$\det G(x) = \det(A + O(x)) \det S(x).$$

Every entry of S is divisible by x, so its determinant is divisible by $x^{\dim K}$, with equality of orders precisely when $\det B \neq 0$. This proves the final equivalence. $\square$

Proposition 5.3 (Two-step inverse principal part). *Let*

$$G(x) = G_0 + xG_1 + x^2 G_2 + O(x^3)$$

be as in Proposition 5.2, and put

$$K_1 = \ker G_0, \qquad \beta_1(u,v) = u^{\mathsf{T}} G_1 v, \qquad K_2 = \ker(\beta_1 \colon K_1 \to K_1^*).$$

For $v \in K_2$, choose $v_1 \in V$ satisfying

$$G_0 v_1 = -G_1 v$$

and define

$$\beta_2(u,v) = u^{\mathsf{T}}(G_2 v + G_1 v_1), \qquad u, v \in K_2.$$

Then v_1 exists, β_2 is independent of its choice, and β_2 is symmetric. Without any nondegeneracy assumption,

$$\#\{i : e_i \geq 1\} = \dim K_1, \qquad \#\{i : e_i \geq 2\} = \dim K_2.$$

β_2 is nondegenerate if and only if all Smith exponents are at most 2. In that case,

$$[x^{-2}]G(x)^{-1} = \iota_2 \beta_2^{-1} \iota_2^*,$$

where $\iota_2 \colon K_2 \hookrightarrow V$.

Proof. Because G_0 is symmetric, $\operatorname{im} G_0$ is the annihilator of K_1. For $v \in K_2$ and $u \in K_1$,

$$u^{\mathsf{T}} G_1 v = \beta_1(u,v) = 0.$$

Thus $G_1 v \in \operatorname{im} G_0$, proving the existence of v_1. Two choices differ by an element $k \in K_1$, and $u^{\mathsf{T}} G_1 k = \beta_1(u,k) = 0$ for $u \in K_2$, proving choice-independence.

Choose a complement to K_1 as in the proof of Proposition 5.2, and first take the Schur complement of its nondegenerate block. This produces a symmetric matrix on K_1 of the form

$$S(x) = xB_1 + x^2 B_2 + O(x^3),$$

where B_1 represents β_1, and the restriction of B_2 to $\ker B_1 = K_2$ is the displayed form β_2. This also proves its symmetry. The positive Smith exponents of G are the Smith exponents of S. Dividing S by x and reducing modulo x gives B_1. The Smith normal form, first for G and then for $x^{-1}S$, therefore gives

$$\#\{i : e_i \geq 1\} = \dim K_1, \qquad \#\{i : e_i \geq 2\} = \dim \ker B_1 = \dim K_2.$$

This count does not require the second crossing form to be nondegenerate.

Now choose a complement E to K_2 in K_1. The restriction of B_1 to E is nondegenerate, and a second Schur complement gives

$$S(x) \sim \begin{pmatrix} x^2 \beta_2 + O(x^3) & 0 \\ 0 & xB_1|_E + O(x^2) \end{pmatrix}.$$

The second block form shows that β_2 is nondegenerate precisely when no Smith exponent exceeds 2. In that case block inversion yields the asserted x^{-2} coefficient. $\square$

Corollary 5.4 (Determinant multiplicity does not determine block pole order)**.** *At a collided Kac weight, the multiplicity of the corresponding factor in* $\det G_N$ *is only the sum of the positive local Smith exponents. The inverse Shapovalov pole order is their maximum.*

Proof. Apply Lemma 5.1 with $x = \Delta - \lambda_n$. $\square$

Lemma 5.5 (Torus form on a singular submodule)**.** *Let* $\chi \in M(c,h)$ *be a singular vector of level* ℓ*, normalized by a fixed PBW coefficient, and let*

$$P_\chi(d) = \langle \chi | V_d(1) | \chi \rangle.$$

For partitions Y, Y' *of the same integer* m*,*

$$\langle \chi | L_Y V_d(1) L_{-Y'} | \chi \rangle = P_\chi(d)\, \rho_m(c, h+\ell, d)_{Y,Y'}.$$

Proof. The assignment

$$L_{-Y}|h+\ell\rangle \longmapsto L_{-Y}\chi$$

is a homomorphism from $M(c, h+\ell)$ to $M(c,h)$, because χ is singular and has L_0-weight $h+\ell$. Pull the torus three-point form on $M(c,h)$ back along this homomorphism in both module arguments. Commuting one Virasoro mode at a time through $V_d(z)$ shows that the pulled-back form obeys the same Ward recursion as $\rho_m(c, h+\ell, d)$. That recursion terminates at the highest-weight matrix element and therefore determines every descendant matrix element uniquely. The pulled-back highest-weight value is $P_\chi(d)$, whereas the normalization $\langle h+\ell | V_d(1) | h+\ell \rangle = 1$ gives highest-weight value 1 for ρ_m. The displayed identity follows. $\square$

Lemma 5.6 (First crossing form on a singular submodule)**.** *Fix* c *and let* $\chi = P|h_0\rangle \in M(c, h_0)$ *be singular of level* ℓ*, where the PBW operator* $P \in U(\mathrm{Vir}_-)$ *is held fixed as* h *varies. Set*

$$b_\chi = \frac{d}{dh} \langle h | \omega(P) P | h \rangle \bigg|_{h=h_0},$$

where $\omega(L_n) = L_{-n}$*. For partitions* Y, Y' *of* m*,*

$$\frac{d}{dh} \langle h | \omega(P) L_Y L_{-Y'} P | h \rangle \bigg|_{h=h_0} = b_\chi\, G_m(c, h_0+\ell)_{Y,Y'}.$$

Proof. Write $S_h(u,v) = \langle h | \omega(u) v | h \rangle$ for the Shapovalov form of $M(c,h)$, and for $u \in U(\mathrm{Vir}_-)$ put $\widetilde{u} = uP \in U(\mathrm{Vir}_-)$. The derivative on the left defines a bilinear form

$$C(u,v) = \frac{d}{dh} S_h(\widetilde{u}, \widetilde{v}) \bigg|_{h=h_0}$$

on the PBW space of $M(c, h_0+\ell)$. At $h = h_0$, the map

$$j_\chi \colon M(c, h_0+\ell) \longrightarrow M(c, h_0), \qquad u|h_0+\ell\rangle \longmapsto u\chi,$$

is a module homomorphism, and its image lies in the radical of the Shapovalov form.

We claim that C is contravariant for the module structure of $M(c, h_0+\ell)$. Contravariance in negative modes, $C(L_{-n}u, v) = C(u, \omega(L_{-n})v)$, holds at every h because the Shapovalov form itself is contravariant and $\widetilde{L_{-n}u} = L_{-n}\widetilde{u}$. For a positive mode, fix $n > 0$ and $v \in U(\mathrm{Vir}_-)$, and let $w_h \in U(\mathrm{Vir}_-)$ be defined by PBW reduction of $L_n v$ in $M(c, h_0+\ell)$, so that $L_n\, v|h_0+\ell\rangle = w_{h_0}|h_0+\ell\rangle$. In $M(c,h)$ set

$$D(h) = L_n \widetilde{v}\, |h\rangle - \widetilde{w_{h_0}}\, |h\rangle.$$

Each coefficient of $D(h)$ in the PBW basis is polynomial in h, and $D(h_0) = 0$ because $\chi = P|h_0\rangle$ is singular and j_χ intertwines the actions at $h = h_0$. Hence $D(h) = (h - h_0)\widetilde{D}(h)$ with $\widetilde{D}$ polynomial in h. Therefore

$$\frac{d}{dh}\Big|_{h_0} S_h(\widetilde{u}, L_n\widetilde{v}) = \frac{d}{dh}\Big|_{h_0} S_h(\widetilde{u}, \widetilde{w_{h_0}}) + S_{h_0}\big(\widetilde{u}, \widetilde{D}(h_0)\big).$$

The last term vanishes because $\widetilde{u}\,|h_0\rangle = j_\chi(u|h_0+\ell\rangle)$ lies in the radical of S_{h_0}. Since $S_h(\widetilde{u}, L_n\widetilde{v}) = S_h(\widetilde{L_{-n}u}, \widetilde{v})$, this proves $C(L_{-n}u, v) = C(u, w_{h_0})$, that is, contravariance in positive modes.

A contravariant form on a Verma module is uniquely determined by its highest-weight value, because PBW commutation recursively reduces every matrix element to that value. Here the highest-weight value is b_χ, so $C = b_\chi G_m(c, h_0 + \ell)$. □

Proposition 5.7 (Earliest Ising collision). *The first repeated Ising Kac weight occurs at level 3:*

$$\Delta_{2,1} = \Delta_{1,3} = \frac{1}{2}, \qquad \ell_{2,1} = 2, \quad \ell_{1,3} = 3.$$

In $M(1/2, 1/2)$, the vectors

$$\xi_2 = \left(L_{-1}^2 - \frac{4}{3}L_{-2}\right)|\tfrac{1}{2}\rangle$$

and

$$\xi_3 = \left(L_{-1}^3 - 3L_{-2}L_{-1} + \frac{3}{4}L_{-3}\right)|\tfrac{1}{2}\rangle$$

are singular. At level 3,

$$\ker G_3(1/2, 1/2) = \operatorname{span}\{L_{-1}\xi_2, \xi_3\},$$

and the local Smith exponents are $(0, 1, 1)$. With the L_{-1}^ℓ-coefficient normalized to 1, their crossing scalars are

$$b_{\xi_2} = \frac{28}{9}, \qquad b_{\xi_3} = -\frac{105}{8},$$

and the level-three crossing form in the ordered basis $(L_{-1}\xi_2, \xi_3)$ is

$$\begin{pmatrix} 140/9 & 0 \\ 0 & -105/8 \end{pmatrix} = \begin{pmatrix} \frac{28}{9}G_1(1/2, 5/2) & 0 \\ 0 & -105/8 \end{pmatrix}.$$

Moreover, the diagonal torus matrix elements are

$$E_2(d) := \langle \xi_2 | V_d(1) | \xi_2 \rangle = \frac{d(d-1)(3d-14)(3d-5)}{9},$$

and

$$E_3(d) := \langle \xi_3 | V_d(1) | \xi_3 \rangle = \frac{d(d-1)(2d-15)(2d-7)(2d-5)(2d-1)}{16}.$$

Proof. The collision statement follows immediately from $\Delta_{r,s} = \lambda_{|4r-3s|}$: the labels of levels at most 2 have distinct values of $|4r - 3s|$, whereas both displayed labels have value 5. Direct commutation with L_1, L_2, L_3 proves that the two vectors are singular. In the ordered basis $(L_{-3}, L_{-2}L_{-1}, L_{-1}^3)|\Delta\rangle$,

$$\det G_3(1/2, \Delta) = 12\Delta^2(2\Delta - 1)^2(3\Delta - 5)(16\Delta - 1).$$

At $\Delta = 1/2$ the matrix has rank one, and

$$L_{-1}\xi_2 = \left(-\frac{4}{3}L_{-3} - \frac{4}{3}L_{-2}L_{-1} + L_{-1}^3\right)|\tfrac{1}{2}\rangle$$

together with ξ_3 spans its radical. The determinant valuation and nullity are both two, so Lemma 5.1 gives the Smith profile. Finally,

$$b_{\xi_j} = \frac{d}{d\Delta}\langle \xi_j|\xi_j\rangle\Big|_{\Delta=1/2}$$

and direct multiplication by $\partial_\Delta G_3|_{\Delta=1/2}$ gives the displayed level-three crossing matrix. The Ward recursion of Lemma 2.4 gives the two displayed torus polynomials. All computations in this proof are exact rational calculations and are replayed by `tools/audit_first_resonances.py`. □

Theorem 5.8 (First fixed-c recursive residue)**.** *At $c = 1/2$, the earliest collided coefficient has the simple principal part*

$$\operatorname*{Res}_{\Delta=1/2} H_3(1/2, \Delta, d) = \frac{9}{28}E_2(d)H_1(1/2, 5/2, d) - \frac{8}{105}E_3(d).$$

Every quantity on the right is evaluated directly at $c = 1/2$.

Proof. The block-diagonal crossing form in Proposition 5.7 has highest-weight inverse scalars $9/28$ and $-8/105$. Proposition 5.2 and Lemma 5.5 therefore give

$$\operatorname*{Res}_{\Delta=1/2} F_2 = \frac{9}{28}E_2, \qquad \operatorname*{Res}_{\Delta=1/2} F_3 = \frac{9}{28}E_2F_1(1/2, 5/2, d) - \frac{8}{105}E_3.$$

The Euler product gives $H_3 = F_3 - F_2 - F_1$; since $G_1(1/2, \Delta) = (2\Delta)$, the coefficient F_1 is regular at $\Delta = \frac{1}{2}$, so $\operatorname{Res}_{\Delta=1/2} H_3 = \operatorname{Res}_{\Delta=1/2}(F_3 - F_2)$. The result follows from $H_1 = F_1 - F_0$. □

Proposition 5.9 (First spin-module collision)**.** *At level 4, the labels $(1, 2)$ and $(2, 2)$ collide at $\Delta = 1/16$. In the PBW basis*

$$(L_{-4},\, L_{-3}L_{-1},\, L_{-2}^2,\, L_{-2}L_{-1}^2,\, L_{-1}^4)|\Delta\rangle,$$

the Shapovalov determinant at $c = 1/2$ is

$$12\Delta^3(2\Delta - 7)(2\Delta - 1)^3(3\Delta - 5)(16\Delta - 21)(16\Delta - 1)^3.$$

At $\Delta = 1/16$, the local Smith exponents are

$$(0, 0, 1, 1, 1).$$

Consequently, $G_4(1/2, \Delta)^{-1}$, $F_4(1/2, \Delta, d)$, and $H_4(1/2, \Delta, d)$ have at most a simple pole at $\Delta = 1/16$.

Proof. Direct Virasoro commutation gives the 5×5 Gram matrix $G_4(1/2, \Delta)$ recorded in Appendix A. Its determinant is the polynomial displayed in the statement. At $\Delta = 1/16$, the upper-left 2×2 minor has determinant 2, while the following three independent vectors lie in the kernel:

$$\begin{pmatrix} -9/16 \\ -3/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -27/64 \\ -9/8 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -465/256 \\ -75/32 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus the specialized matrix has rank 2 and nullity 3. The determinant has order 3 at $1/16$, so Lemma 5.1 gives the stated Smith exponents and inverse pole bound. The definitions of F_4 and H_4 then give the same bound for the block coefficients. □

Proposition 5.10 (Singular data at the first spin collision). *Put $h_0 = 1/16$. The following vectors in $M(1/2, h_0)$ are singular:*

$$\chi_2 = \left(L_{-1}^2 - \frac{3}{4}L_{-2}\right)|h_0\rangle$$

and

$$\chi_4 = \left(L_{-1}^4 - \frac{25}{6}L_{-2}L_{-1}^2 + \frac{49}{144}L_{-2}^2 + \frac{11}{6}L_{-3}L_{-1} - \frac{1}{4}L_{-4}\right)|h_0\rangle.$$

At level 4,

$$\ker G_4(1/2, h_0) = \operatorname{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

In this ordered basis the crossing form of Proposition 5.2, with $x = \Delta - h_0$, is

$$\beta_4 = \begin{pmatrix} -119/8 & -693/32 & 0 \\ -693/32 & -9471/128 & 0 \\ 0 & 0 & 13475/648 \end{pmatrix} = \begin{pmatrix} -\frac{7}{4}G_2(1/2, 33/16) & 0 \\ 0 & 13475/648 \end{pmatrix}.$$

Finally, the two highest-weight torus matrix elements are

$$P_2(d) := \langle\chi_2|V_d(1)|\chi_2\rangle = \frac{d(d-1)(2d-7)(2d-1)}{4}$$

and

$$\begin{aligned} P_4(d) &:= \langle\chi_4|V_d(1)|\chi_4\rangle \\ &= \frac{d(d-1)(2d-7)(2d-1)(3d-14)(3d-5)(6d-55)(6d-1)}{1296}. \end{aligned}$$

Proof. Apply $L_1, \dots, L_\ell$ to the displayed level-ℓ vector and use the Virasoro commutator. Collecting coefficients in the PBW bases at levels $\ell - 1, \dots, 0$ gives zero for χ_2 and for χ_4; modes larger than ℓ vanish by degree. Thus both vectors are singular.

In the level-four PBW basis of Proposition 5.9, the three asserted kernel vectors are the columns

$$\begin{pmatrix} 0 \\ 0 \\ -3/4 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -3/2 \\ -3/2 \\ 0 \\ -3/4 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} -1/4 \\ 11/6 \\ 49/144 \\ -25/6 \\ 1 \end{pmatrix}.$$

Multiplication by the matrix $G_4(1/2, h_0)$ of Appendix A gives zero. They are independent, and Proposition 5.9 gives nullity three, proving the kernel identity.

Multiplying these columns on both sides of $\partial_\Delta G_4(1/2, \Delta)|_{\Delta = h_0}$ gives the displayed crossing matrix. Direct level-two commutation gives

$$G_2(1/2, 33/16) = \begin{pmatrix} 17/2 & 99/8 \\ 99/8 & 1353/32 \end{pmatrix},$$

which verifies the second expression for β_4. Finally, repeated use of the primary-field Ward commutator in the proof of Lemma 2.4, followed by collection and factorization over $\mathbb{Q}[d]$, gives the displayed P_2 and P_4. □

Theorem 5.11 (Spin-sector fixed-c recursive principal part). *At $c = 1/2$, the level-four reduced coefficient satisfies the finite fixed-central-charge recursion*

$$\operatorname*{Res}_{\Delta = 1/16} H_4(1/2, \Delta, d) = -\frac{4}{7}P_2(d)H_2(1/2, 33/16, d) + \frac{648}{13475}P_4(d).$$

Equivalently, this residue is

$$\frac{d(d-1)(2d-7)(2d-1)}{40425}\left(248d^4 - 5752d^3 + 25274d^2 - 33630d + 5775\right).$$

Every quantity on the right is evaluated directly at $c = 1/2$.

Proof. At levels 2, 3, and 4, the radicals at $\Delta = h_0$ are respectively

$$\operatorname{span}\{\chi_2\}, \qquad \operatorname{span}\{L_{-1}\chi_2\}, \qquad \operatorname{span}\{L_{-2}\chi_2, L_{-1}^2\chi_2, \chi_4\}.$$

The crossing forms on the first two spaces are

$$-\frac{7}{4}, \qquad -\frac{7}{4}G_1(1/2, 33/16),$$

and the third is the form in Proposition 5.10. These identities follow by differentiating the level-two, level-three, and level-four Gram matrices; all three crossing forms are nondegenerate. Apply Proposition 5.2, then use Lemma 5.5 on each singular summand. Taking the trace against the corresponding torus matrix gives

$$\begin{aligned}
\operatorname*{Res}_{\Delta=h_0} F_2 &= -\frac{4}{7}P_2, \\
\operatorname*{Res}_{\Delta=h_0} F_3 &= -\frac{4}{7}P_2F_1(1/2, 33/16, d), \\
\operatorname*{Res}_{\Delta=h_0} F_4 &= -\frac{4}{7}P_2F_2(1/2, 33/16, d) + \frac{648}{13475}P_4.
\end{aligned}$$

The Euler product has coefficients $1, -1, -1, 0, 0$ through degree four. Hence $H_4 = F_4 - F_3 - F_2$, while $H_2 = F_2 - F_1 - F_0$. Substitution gives the recursive formula. Inserting the explicit polynomials from Proposition 5.10 and the level-two coefficient from the inverse Shapovalov matrix gives the second displayed expression. □

6 Generic residue data and the first nontransverse diamond

Definition 6.1 (Generic residue data used for algebraic confluence)**.** For $b \in \mathbb{C}^\times$, set

$$c(b) = 1 + 6(b + b^{-1})^2, \qquad h_{r,s}(b) = \frac{(b + b^{-1})^2 - (rb + sb^{-1})^2}{4}.$$

Throughout the confluence calculations,

$$b_0 = \frac{2i}{\sqrt{3}}, \qquad c(b_0) = \frac{1}{2}.$$

Put $\Lambda^2 = (b + b^{-1})^2 - 4d$, and define

$$\begin{aligned}
\mathcal{P}_{r,s}(b,d) = \prod_{\substack{1\le k\le 2r-1\\ k \text{ odd}}} \prod_{\substack{1\le l\le 2s-1\\ l \text{ odd}}} &\frac{\Lambda^2 - (kb + lb^{-1})^2}{4} \\
&\times \frac{\Lambda^2 - (kb - lb^{-1})^2}{4},
\end{aligned}$$

and

$$\mathcal{A}_{r,s}(b) = \frac{1}{2} \prod_{\substack{1-r\le p\le r,\ 1-s\le q\le s\\ (p,q)\ne(0,0),(r,s)}} (pb + qb^{-1})^{-1}, \qquad \mathcal{R}_{r,s}(b,d) = \mathcal{A}_{r,s}(b)\mathcal{P}_{r,s}(b,d).$$

For generic b, $\mathcal{R}_{r,s}$ is the residue coefficient in the eta-reduced torus recursion, stated as Proposition 6.4 below.

Remark 6.2 (Source, normalization, and imported identities)**.** The formula is equations (7)–(9) of [8, Section 3]; the recursion was conjectured in [5]. The product $\mathcal{P}_{r,s}$ is the product of the two torus fusion polynomials of [8], written without choosing a square root of Λ^2. In that normalization the singular vector has L_{-1}^{rs}-coefficient 1. Two identities at generic b are imported from that literature rather than rederived here. First, with $\chi_{r,s}(b)$ the normalized singular vector and b_χ the crossing scalar of Lemma 5.6,

$$b_{\chi_{r,s}(b)} = \mathcal{A}_{r,s}(b)^{-1}; \tag{6.1}$$

this is the classical norm-derivative formula behind Zamolodchikov-type residues [2, 8]. Second,

$$\langle \chi_{r,s}(b) | V_d(1) | \chi_{r,s}(b) \rangle = \mathcal{P}_{r,s}(b, d); \tag{6.2}$$

this is the content of the residue computation in [8, Section 3] and is *not* a one-line consequence of the Ward recursion. Both sides of each identity are rational in b, so the identities continue to every specialization at which the normalized singular vector is regular. This fixes the normalization used below, rather than merely identifying the polynomials up to a scalar. Both identities are verified in exact arithmetic by the audit scripts of Remark 7.5: (6.2) directly at b_0 for the labels $(2,1), (1,3), (1,5), (4,1), (1,2), (2,2)$, at a generic sample point for $(2,1), (1,2)$, and, for $(4,2), (1,6)$, through the level-eight and level-six residues checked by `tools/audit_grade10_direct.py`; and (6.1) at b_0 for the labels $(2,1), (1,3), (1,2), (2,2)$ through the crossing scalars of Section 5.

Lemma 6.3 (Distinct weights and no shifted self-poles at generic b)**.** *For positive labels (r,s) and (u,v),*

$$h_{r,s}(b) + rs - h_{u,v}(b)$$

is not the zero rational function of b. Moreover, for distinct positive labels $(r,s) \neq (u,v)$, the difference $h_{r,s}(b) - h_{u,v}(b)$ is not the zero rational function.

Proof. The Kac formula gives the identity

$$h_{r,s}(b) + rs = h_{r,-s}(b).$$

If this were identically equal to $h_{u,v}(b)$, then

$$(rb - sb^{-1})^2 = (ub + vb^{-1})^2$$

in $\mathbb{C}(b)$. Since this is an integral domain,

$$rb - sb^{-1} = \pm(ub + vb^{-1}).$$

Comparison of the coefficients of b and b^{-1} contradicts $r, s, u, v > 0$. For the second assertion, the same argument gives $rb + sb^{-1} = \pm(ub + vb^{-1})$, and positivity forces $(r,s) = (u,v)$. $\square$

Proposition 6.4 (Generic eta-reduced recursion)**.** *Fix $N \geq 1$. For all b outside a finite set of algebraic values,*

$$H_N(c(b), \Delta, d) = \sum_{\substack{a=(r,s)\in\mathbb{Z}^2_{>0} \\ \ell_a = rs \leq N}} \frac{\mathcal{R}_{r,s}(b,d)}{\Delta - h_{r,s}(b)} H_{N-\ell_a}\big(c(b), h_{r,s}(b) + rs, d\big)$$

as an identity of rational functions of Δ, polynomial in d.

Proof. Given the two imported identities of Remark 6.2, the proposition follows from the results of Sections 4 and 5. By Lemma 6.3, outside finitely many values of b the weights $h_{r,s}(b)$ with $rs \le N$ are pairwise distinct, and no shifted weight $z_{r,s} = h_{r,s}(b) + rs$ is a degenerate weight of level at most N.

Fix such a b, write $c = c(b)$, and fix a label $a = (r, s)$ with $rs \le N$. By distinctness, $h_{r,s}(b)$ lies on no other Kac divisor of level at most N, so the grade-N Kac determinant has valuation exactly $p(N - rs)$ at $\Delta = h_{r,s}(b)$. The grade-N image of the injective singular embedding $j_{\chi_{r,s}(b)}$ is a $p(N - rs)$-dimensional subspace of the kernel of the specialized Gram matrix. By Lemma 5.1 the nullity is at most the valuation, so the kernel is exactly that image, and every positive local Smith exponent equals 1; hence Proposition 5.2 applies. By Lemma 5.6, the crossing form on this kernel is $b_\chi \, G_{N-rs}(c, z_{r,s})$, which is nondegenerate because $z_{r,s}$ is not a degenerate weight and, by (6.1), $b_\chi^{-1} = \mathcal{A}_{r,s}(b) \neq 0$. Taking the trace of the residue of G_N^{-1} against ρ_N and factoring the pulled-back torus form by Lemma 5.5 and (6.2) gives

$$\operatorname*{Res}_{\Delta=h_{r,s}(b)} F_N(c, \Delta, d) = \mathcal{A}_{r,s}(b)\, \mathcal{P}_{r,s}(b, d)\, F_{N-rs}(c, z_{r,s}, d) = \mathcal{R}_{r,s}(b, d)\, F_{N-rs}(c, z_{r,s}, d).$$

The difference between F_N and the sum of these principal parts has no finite pole, hence is a polynomial in Δ; it is bounded as $\Delta \to \infty$ by Lemma 4.1, hence equals the constant $p(N)$:

$$F_N(c, \Delta, d) = p(N) + \sum_{\ell_a \le N} \frac{\mathcal{R}_a(b, d)}{\Delta - h_a(b)} F_{N-\ell_a}(c, z_a, d).$$

Finally, multiply by the Euler coefficients e_j and sum. The generating identity in the proof of Lemma 4.1 gives $\sum_j e_j p(N - j) = \delta_{N,0}$, while the residue sums reassemble into shifted H-coefficients:

$$H_N(c, \Delta, d) = \delta_{N,0} + \sum_{\ell_a \le N} \frac{\mathcal{R}_a(b, d)}{\Delta - h_a(b)} H_{N-\ell_a}(c, z_a, d).$$

For $N \ge 1$ this is the displayed recursion. □

Proposition 6.5 (First nontransverse Smith layer)**.** *Let $h_E = 1/2$, and let U_2, U_3 be the submodules generated by ξ_2, ξ_3 in Proposition 5.7. Their first intersection is generated at total level 7 by a normalized singular vector ξ_7. More explicitly, the normalized child singular vectors are*

$$\begin{aligned}\phi_5 = \big(&L_{-1}^5 - 15L_{-2}L_{-1}^3 + 36L_{-2}^2L_{-1} + \tfrac{99}{4}L_{-3}L_{-1}^2 \\ &- 45L_{-3}L_{-2} - 54L_{-4}L_{-1} + 45L_{-5}\big)|\tfrac{5}{2}\rangle,\end{aligned}$$

$$\phi_4 = \big(L_{-1}^4 - \tfrac{40}{3}L_{-2}L_{-1}^2 + 16L_{-2}^2 + \tfrac{88}{3}L_{-3}L_{-1} - \tfrac{152}{3}L_{-4}\big)|\tfrac{7}{2}\rangle,$$

and

$$\xi_7 = j_{\xi_2}(\phi_5) = j_{\xi_3}(\phi_4).$$

The 15 *local Smith exponents of $G_7(1/2, h_E + x)$ are*

$$(0^4, 1^{10}, 2).$$

For the second crossing form of Proposition 5.3, normalized by the displayed ξ_7,

$$\gamma_7 = -700700.$$

Proof. The two child highest weights are

$$h_E + 2 = \frac{5}{2} = \lambda_{11}, \qquad h_E + 3 = \frac{7}{2} = \lambda_{13}.$$

Their first singular vectors have relative levels 5 and 4, with labels $(1, 5)$ and $(4, 1)$, respectively. Direct Virasoro commutation checks the two displayed vectors. Their images are nonzero singular vectors at total level 7; uniqueness and their common L_{-1}^7-coefficient 1 prove the displayed equality. The embedding diagram in [7, Proposition 5.4 and Figure 5.4] also shows that there is no earlier intersection and that the radical is the sum of the two singular submodules.

At grade 7, the radical therefore has dimension

$$p(5) + p(4) - 1 = 7 + 5 - 1 = 11.$$

The only Kac labels of weight 1/2 and level at most 7 are $(2, 1)$ and $(1, 3)$, so the determinant valuation is

$$p(5) + p(4) = 12.$$

The Smith exponents are consequently $(0^4, 1^{10}, 2)$.

For completeness, the second crossing scalar can be obtained without an off-critical deformation. In the level-seven PBW basis, form

$$G_7(1/2, h_E + x) = G_0 + xG_1 + x^2G_2 + O(x^3),$$

solve $G_0v_1 = -G_1\xi_7$, and evaluate

$$\xi_7^{\mathsf{T}}(G_2\xi_7 + G_1v_1).$$

Exact PBW commutation gives -700700. Choice-independence follows from Proposition 5.3. This 15×15 fixed-c calculation is replayed independently in `tools/audit_first_resonances.py`. □

Proposition 6.6 (No earlier nontransverse pole)**.** *For every $N < 7$ and every finite internal weight, all local Smith exponents of $G_N(1/2, \Delta)$ are at most 1. Consequently G_N^{-1}, F_N, and H_N have at most simple poles in Δ. Together with Proposition 6.5, this proves that the double pole at $\Delta = 1/2$ and grade 7 is the first double pole of the Ising Verma block.*

Proof. For a Kac weight represented by only one label $a = (r, s)$ with $\ell_a \le N$, injectivity of the singular-vector embedding gives a radical of dimension $p(N - \ell_a)$. This equals the valuation contributed by that label to the Kac determinant. Lemma 5.1 therefore makes all corresponding positive Smith exponents equal to 1.

It remains to inspect repeated labels. Proposition 3.2, or direct enumeration of $rs \le 6$, shows that the complete list of repeated classes below grade 7 is

weight	labels	singular levels
$\lambda_5 = \frac{1}{2}$	$(2, 1), (1, 3)$	$2, 3$
$\lambda_2 = \frac{1}{16}$	$(1, 2), (2, 2)$	$2, 4$
$\lambda_1 = 0$	$(1, 1), (2, 3)$	$1, 6.$

The two energy branches have no intersection below grade 7 by Proposition 6.5.

For the spin class, the child highest weights along the level-two and level-four branches are 33/16 and 65/16. Their Kac determinants have no zero below relative levels 8 and 6, respectively. A first intersection below grade 7 would be singular; injectivity would pull it back along the level-two branch to a singular vector of relative level at most 4, which is impossible.

For the vacuum class, the level-one branch is generated by $L_{-1}|0\rangle$ and has child highest weight 1. The Kac determinant of $M(1/2, 1)$ has no zero through relative level 5. Hence the level-six singular vector cannot lie in the level-one branch, since its preimage would be singular at relative level 5.

Thus, at every repeated weight and every $N < 7$, the relevant singular submodules have zero intersection. The radical dimension is therefore the sum $\sum_a p(N - \ell_a)$, exactly the Kac-determinant valuation. A final application of Lemma 5.1 proves the Smith and pole bounds. □

Theorem 6.7 (First nontransverse fixed-c principal part)**.** *Define*

$$E_5(d) = \mathcal{P}_{1,5}(b_0, d), \qquad E_4(d) = \mathcal{P}_{4,1}(b_0, d).$$

Explicitly,

$$E_5(d) = \frac{1}{16}d(d-20)(d-13)(d-11)(d-6)(d-1)(2d-15)(2d-7)(2d-5)(2d-1),$$

$$E_4(d) = \frac{1}{9}d(d-20)(d-13)(d-11)(d-6)(d-1)(3d-14)(3d-5),$$

and $E_2E_5 = E_3E_4$. The subscripts of E_5 and E_4 record the relative levels of the child singular vectors ϕ_5 and ϕ_4 of Proposition 6.5, in parallel with the parent-module elements E_2, E_3 of Proposition 5.7. Put

$$Q_5^E(d) = \frac{\partial}{\partial z}\{(z-\tfrac{5}{2})H_5(1/2, z, d)\}\Big|_{z=5/2},$$

$$Q_4^E(d) = \frac{\partial}{\partial z}\{(z-\tfrac{7}{2})H_4(1/2, z, d)\}\Big|_{z=7/2}.$$

These are lower-grade finite parts. They can be obtained directly from the fixed-c inverse-Shapovalov matrices, or recursively from Theorem 8.6. For

$$\alpha = (2,1), \quad \beta = (1,3), \quad \mu = (1,5), \quad \nu = (4,1),$$

let

$$\delta_\alpha = h_\alpha + 2 - h_\mu, \qquad \delta_\beta = h_\beta + 3 - h_\nu,$$

and, with dots denoting d/db at b_0, define the explicit finite polynomial

$$\Theta_7(d) = \sum_{(\kappa,\tau)=(\alpha,\mu),(\beta,\nu)} \left(\frac{\dot{\mathcal{R}}_\kappa \mathcal{R}_\tau + \mathcal{R}_\kappa \dot{\mathcal{R}}_\tau}{\dot\delta_\kappa} - \frac{\mathcal{R}_\kappa \mathcal{R}_\tau \ddot\delta_\kappa}{2\dot\delta_\kappa^2} \right)_{b=b_0}.$$

Explicitly,

$$\begin{aligned}\Theta_7(d) = &-\frac{d^2(d-1)^2(2d-15)(2d-7)(2d-5)(2d-1)(3d-14)(3d-5)}{2164322160000} \\ &\times \big(84001d^4 - 3376370d^3 + 44478887d^2 \\ &\qquad - 209562838d + 205096320\big).\end{aligned}$$

Then

$$\begin{aligned}\mathrm{PP}_{\Delta=1/2}\, H_7(1/2, \Delta, d) = &-\frac{E_2(d)E_5(d)}{700700\,(\Delta-\frac{1}{2})^2} \\ &+ \frac{\frac{9}{28}E_2(d)Q_5^E(d) - \frac{8}{105}E_3(d)Q_4^E(d) + \Theta_7(d)}{\Delta-\frac{1}{2}}.\end{aligned}$$

This is the earliest double pole of the Ising block.

Proof. Two remarks fix the logical status of the argument. First, the identification of the confluent Laurent expansion below with the fixed-c Laurent coefficients of H_7 is the grade-seven instance of the rational-continuation argument proved in Theorem 8.6, exactly as invoked for grade ten in Theorem 7.3; nothing in the proof of Theorem 8.6 depends on the present theorem. Second, the double-pole coefficient also has an intrinsic derivation that avoids the confluence altogether: Propositions 5.3 and 6.5 give $[x^{-2}]G_7(1/2, \frac{1}{2}+x)^{-1} = \xi_7\xi_7^{\mathsf{T}}/\gamma_7$, Lemma 5.5 applied along either branch of the diamond gives $\langle \xi_7 | V_d(1) | \xi_7 \rangle = E_2E_5 = E_3E_4$, and multiplication by the Euler product does not affect the x^{-2} coefficient at grade seven because all lower grades have at most simple poles by Proposition 6.6.

Equation (6.2) gives the four fusion polynomials. Direct evaluation of the residue products and Kac slopes gives

(r,s)	$(2,1)$	$(1,3)$	$(1,5)$	$(4,1)$
$\mathcal{A}_{r,s}(b_0)$	$9/28$	$-8/105$	$-1/102375$	$3/61600$

and

$$\frac{\dot{h}_{2,1}-\dot{h}_{1,3}}{\dot{h}_{2,1}-\dot{h}_{1,5}}=\frac{5}{11},\qquad \frac{\dot{h}_{1,3}-\dot{h}_{2,1}}{\dot{h}_{1,3}-\dot{h}_{4,1}}=\frac{5}{13}.$$

Either path therefore expresses the coefficient produced jointly by their cancelled t^{-1} parts:

$$\frac{5}{11}\mathcal{A}_{2,1}\mathcal{A}_{1,5}=\frac{5}{13}\mathcal{A}_{1,3}\mathcal{A}_{4,1}=-\frac{1}{700700}.$$

This agrees with the intrinsic inverse second crossing scalar γ_7^{-1} from Proposition 6.5.

Expanding the two outer recursion terms at $\Delta=1/2$, their cancelled t^{-1} numerators give the displayed double pole. Their finite parts are respectively

$$\frac{9}{28}E_2Q_5^E+T_{\alpha,\mu},\qquad -\frac{8}{105}E_3Q_4^E+T_{\beta,\nu},$$

where the two T's are precisely the summands defining Θ_7. Differentiating the finite products of Definition 6.1 and factoring over $\mathbb{Q}[d]$ gives the displayed explicit polynomial form of Θ_7. This proves the complete principal part. Its global minimality in grade is Proposition 6.6. □

7 The spin-module grade-ten diamond

Proposition 7.1 (Grade-ten Smith profile at $\Delta=1/16$). *Let $h_0=1/16$, and let*

$$U_2=U(\mathrm{Vir}_-)\chi_2,\qquad U_4=U(\mathrm{Vir}_-)\chi_4$$

be the two singular submodules of $M(1/2,h_0)$ from Proposition 5.10. Their first intersection is generated at total level 10 by a singular vector χ_{10}, and

$$\dim\ker G_{10}(1/2,h_0)=32.$$

The 42 local Smith exponents of $G_{10}(1/2,h_0+x)$ are

$$\left(0^{10},1^{31},2\right).$$

In particular, the inverse Shapovalov matrix has a nonzero rank-one coefficient at order x^{-2}.

Proof. The weight h_0 is the interior Ising weight labelled by $(1,2)$, so it is in Class R^+, Case 1+, in the classification of [7, Section 5.1.4]. The commutative embedding diagram in [7, Proposition 5.4 and Figure 5.4] describes the two branches and their successive intersections. The identification of the first Jantzen layer with their sum, hence with the maximal proper submodule, is [7, Theorem 6.3(i) and equation (6.3)].

The two highest weights are

$$h_0+2=\frac{33}{16}=\lambda_{10},\qquad h_0+4=\frac{65}{16}=\lambda_{14}.$$

The first singular vector inside U_2 has relative level 8, with Kac label $(4,2)$, and the first singular vector inside U_4 has relative level 6, with Kac label $(1,6)$. Both therefore have total level 10 and weight

$$h_0+10=\frac{161}{16}=\lambda_{22}.$$

The embeddings are injective by [7, the paragraph preceding Corollary 5.2]. Their two images are proportional by uniqueness of a singular vector at a fixed level, [7, Proposition 5.1]. Hence the level-ten intersection is one dimensional. There is no earlier intersection: a vector at the first intersection grade would be singular, whereas the Kac determinant for $M(1/2, 33/16)$, respectively $M(1/2, 65/16)$, has no zero below relative level 8, respectively 6.

The grade-ten radical is the grade-ten part of $U_2 + U_4$. Since

$$p(8) = 22, \qquad p(6) = 11,$$

and the intersection has dimension one at this grade,

$$\dim\ker G_{10}(1/2, h_0) = 22 + 11 - 1 = 32.$$

By Proposition 3.2, the only positive Kac labels with $\Delta_{r,s} = h_0$ and $rs \leq 10$ are $(1, 2)$ and $(2, 2)$. The Kac determinant formula therefore gives

$$\operatorname{ord}_x \det G_{10}(1/2, h_0 + x) = p(8) + p(6) = 33.$$

There are $p(10) = 42$ Smith exponents, exactly 32 of them are positive, and their sum is 33. Thus 31 positive exponents equal 1, one equals 2, and the remaining 10 vanish. Lemma 5.1 gives the final assertion. □

Lemma 7.2 (Exact constants of the grade-ten confluence)**.** *Let $b_0 = 2i/\sqrt{3}$, so $c(b_0) = 1/2$. Then*

(r, s)	$(1, 2)$	$(2, 2)$	$(4, 2)$	$(1, 6)$
$\mathcal{A}_{r,s}(b_0)$	$-4/7$	$648/13475$	$48/32035128125$	$-8/320945625.$

If a dot denotes d/db at b_0, then

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}} = \frac{1}{5}, \qquad \frac{\dot{h}_{2,2} - \dot{h}_{1,2}}{\dot{h}_{2,2} - \dot{h}_{1,6}} = \frac{1}{7},$$

and

$$\frac{1}{5}\mathcal{A}_{1,2}\mathcal{A}_{4,2} = \frac{1}{7}\mathcal{A}_{2,2}\mathcal{A}_{1,6} = -\frac{192}{1121229484375}.$$

Moreover,

$$\begin{aligned}
\mathcal{P}_{4,2}(b_0, d) = {} & \frac{1}{20736} d(d-20)(d-13)(d-11)(d-6)(d-1) \\
& \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1) \\
& \times (3d-14)(3d-5)(6d-55)(6d-1), \\
\mathcal{P}_{1,6}(b_0, d) = {} & \frac{1}{64} d(d-20)(d-13)(d-11)(d-6)(d-1) \\
& \times (2d-57)(2d-35)(2d-15)(2d-7)(2d-5)(2d-1),
\end{aligned}$$

and

$$P_2(d)\mathcal{P}_{4,2}(b_0, d) = P_4(d)\mathcal{P}_{1,6}(b_0, d).$$

Proof. Substitute $b_0 = 2i/\sqrt{3}$ in the finite products of Definition 6.1. Direct differentiation gives

$$c(b_0) = \frac{1}{2}, \qquad c'(b_0) = \frac{7i\sqrt{3}}{2} \neq 0,$$

so $b - b_0$ is a valid local deformation parameter. The Kac-weight derivatives are

(r, s)	$(1, 2)$	$(2, 2)$	$(4, 2)$	$(1, 6)$
$16\dot{h}_{r,s}/(i\sqrt{3})$	9	-7	-71	105.

These four values give the two slope ratios. Multiplication of the four displayed $\mathcal{A}$-values gives the common nested residue. Factoring the two $\mathcal{P}$-products over $\mathbb{Q}[d]$ gives the last three identities. □

Theorem 7.3 (Leading double-pole coefficient at grade ten)**.** *Normalize all singular vectors by setting their L_{-1}^{ℓ} coefficient equal to 1. Let $\psi_8 \in M(1/2, 33/16)$ and $\psi_6 \in M(1/2, 65/16)$ be the singular vectors of levels 8 and 6, respectively. Then*

$$\chi_{10} = j_{\chi_2}(\psi_8) = j_{\chi_4}(\psi_6).$$

Define

$$P_8^{(33/16)}(d) = \langle \psi_8 | V_d(1) | \psi_8 \rangle, \qquad P_6^{(65/16)}(d) = \langle \psi_6 | V_d(1) | \psi_6 \rangle.$$

They are the explicit polynomials

$$P_8^{(33/16)}(d) = \mathcal{P}_{4,2}(b_0, d), \qquad P_6^{(65/16)}(d) = \mathcal{P}_{1,6}(b_0, d)$$

listed in Lemma 7.2. Write

$$G_{10}(1/2, h_0 + x) = G_0 + xG_1 + x^2 G_2 + O(x^3),$$

choose v_1 with

$$G_0 v_1 = -G_1 \chi_{10},$$

and set

$$\gamma_{10} = \chi_{10}^{\mathsf{T}} (G_2 \chi_{10} + G_1 v_1).$$

Then it is independent of the choice of v_1, and

$$\gamma_{10} = -\frac{1121229484375}{192}.$$

Consequently,

$$\lim_{\Delta \to h_0} (\Delta - h_0)^2 H_{10}(1/2, \Delta, d) = \frac{P_2(d) P_8^{(33/16)}(d)}{\gamma_{10}} = \frac{P_4(d) P_6^{(65/16)}(d)}{\gamma_{10}}.$$

This is an intrinsic fixed-c expression for the spin-module nontransverse Laurent coefficient.

Proof. The two compositions in the first display are nonzero singular vectors of total level 10. Proposition 5.1 of [7] makes them proportional. Their L_{-1}^{10} coefficients are both 1, so they are equal. The Kac determinants of the child modules of highest weights 33/16 and 65/16 have no zero below levels 8 and 6, respectively. Consequently the standard L_{-1}^{ℓ}-normalized singular-vector recursion has a nonsingular coefficient system at every preceding grade. Cramer's rule then shows that the normalized rational families $\psi_8(b)$ and $\psi_6(b)$ are regular at b_0. Hence the rational identity (6.2) specializes to

$$P_8^{(33/16)} = \mathcal{P}_{4,2}(b_0, d), \qquad P_6^{(65/16)} = \mathcal{P}_{1,6}(b_0, d).$$

Let $K_1 = \ker G_0 = (U_2 + U_4)_{10}$, and let β_1 be its first crossing form. By Lemma 5.6, the restriction of β_1 to U_2 is the scalar b_{χ_2} times the Shapovalov form on $M(1/2, 33/16)$. Since ψ_8 is singular, $\beta_1(\chi_{10}, U_2) = 0$. The same argument through U_4 gives $\beta_1(\chi_{10}, U_4) = 0$. Therefore $\chi_{10} \in K_2 = \ker \beta_1$.

Proposition 7.1 shows that exactly one Smith exponent is at least 2. The unconditional Smith-layer count in Proposition 5.3 therefore shows that K_2 is spanned by χ_{10}. Since that exponent is exactly 2, the same proposition shows that its second crossing form is the nonzero scalar γ_{10}, and gives

$$[x^{-2}] G_{10}(1/2, h_0 + x)^{-1} = \frac{\chi_{10} \chi_{10}^{\mathsf{T}}}{\gamma_{10}}.$$

Taking the trace against the specialized torus matrix yields

$$[x^{-2}]F_{10}(1/2, h_0 + x, d) = \frac{\langle \chi_{10}|V_d(1)|\chi_{10}\rangle}{\gamma_{10}}.$$

For $N < 10$, the two singular submodules have zero intersection at grade N. Thus the radical dimension is

$$p(N-2) + p(N-4),$$

with $p(k) = 0$ for $k < 0$, and this equals the Kac-determinant valuation. Lemma 5.1 therefore shows that all lower-level Smith exponents at h_0 are at most one. Multiplication by the Euler product consequently does not alter the x^{-2} coefficient at grade 10. Finally, Lemma 5.5, applied along either side of the embedding diamond, gives

$$\langle \chi_{10}|V_d(1)|\chi_{10}\rangle = P_2(d)P_8^{(33/16)}(d) = P_4(d)P_6^{(65/16)}(d).$$

The Gram matrices and the normalized singular-vector coefficients are rational at $c = 1/2$, so the defining linear system gives $\gamma_{10} \in \mathbb{Q}$. It remains to evaluate this intrinsic scalar. Apply the generic recursion of Definition 6.1 algebraically near b_0. This comparison with the fixed-c coefficient is justified directly by rational continuation: for Δ away from the finite grade-ten Kac set, the inverse-Shapovalov coefficient is regular in $t = b - b_0$ and equals the generic recursion on the punctured neighborhood. After the resonant paths are combined, their constant term is therefore the specialized coefficient on that open set. Both sides are rational in Δ, so the equality, including the coefficient of $(\Delta - h_0)^{-2}$, holds identically. This is the grade-ten instance of the general argument in Theorem 8.6 and does not use the value of γ_{10}.

The two nested paths have opposite t^{-1} numerator coefficients. After their cancellation, the resulting double-pole coefficient can be written using the $(1, 2) \to (4, 2)$ path as

$$\frac{\dot{h}_{1,2} - \dot{h}_{2,2}}{\dot{h}_{1,2} - \dot{h}_{4,2}}\,\mathcal{A}_{1,2}(b_0)\mathcal{A}_{4,2}(b_0)P_2(d)P_8^{(33/16)}(d).$$

Lemma 7.2 evaluates this scalar coefficient as $-192/1121229484375$. Rewriting the same combined coefficient using the other path gives the equal expression with slope ratio $1/7$; the two expressions are alternatives, not additive contributions. Comparison with the already proved intrinsic coefficient P_2P_8/γ_{10} yields the displayed value of γ_{10}. □

Theorem 7.4 (Finite recursive grade-ten principal part)**.** *Define the finite shifted-block parts*

$$Q_8(d) = \frac{\partial}{\partial z}\left\{(z - \tfrac{33}{16})H_8(1/2, z, d)\right\}\Big|_{z=33/16}$$

and

$$Q_6(d) = \frac{\partial}{\partial z}\left\{(z - \tfrac{65}{16})H_6(1/2, z, d)\right\}\Big|_{z=65/16}.$$

These lower-grade finite parts are computable from the fixed-c inverse-Shapovalov matrices. Alternatively, one may use the finite recursion in Theorem 8.6. Their definitions require no off-critical numerical limit. Put

$$\begin{aligned}\Theta_{10}(d) = -\,&\frac{d^2(d-1)^2(2d-7)^2(2d-1)^2(3d-14)(3d-5)(6d-55)(6d-1)}{55273527956146500000}\\ &\times \big(4246404 64d^8 - 41506307104d^7 + 1664502727528d^6\\ &\quad - 35552058801520d^5 + 438148123547677d^4 - 3142854705332170d^3\\ &\quad + 12488401942129875d^2 - 23870935242348750d + 14566042731240000\big).\end{aligned}$$

Then the entire principal part of the level-ten reduced coefficient at $\Delta = h_0 = 1/16$ is

$$\begin{aligned}\mathrm{PP}_{\Delta=h_0} H_{10}(1/2,\Delta,d) = &-\frac{192\,P_2(d)P_8^{(33/16)}(d)}{1121229484375\,(\Delta-h_0)^2}\\ &+\frac{-\frac{4}{7}P_2(d)Q_8(d)+\frac{648}{13475}P_4(d)Q_6(d)+\Theta_{10}(d)}{\Delta-h_0}.\end{aligned}$$

Every term on the right is finite and is evaluated at $c = 1/2$. Thus this is a complete finite recursive formula for the principal part. Closed forms of the finite-part polynomials Q_8 and Q_6 (of degrees 16 and 12 in d, with unwieldy rational coefficients) are recorded in machine-readable form, and verified against the inverse-Shapovalov definition, in `tools/audit_grade10_direct.py`*; see Remark 7.5.*

Proof. Let $t = b - b_0$, and abbreviate the four labels

$$\alpha = (1,2), \quad \beta = (2,2), \quad \mu = (4,2), \quad \nu = (1,6).$$

Set

$$z_\alpha = h_\alpha + 2, \qquad z_\beta = h_\beta + 4, \qquad \delta_\alpha = z_\alpha - h_\mu, \qquad \delta_\beta = z_\beta - h_\nu.$$

Only the α- and β-terms of the generic recursion can develop a pole at $\Delta = h_0$. Their numerators at grade ten are

$$C_\alpha(b) = \mathcal{R}_\alpha(b,d)H_8(c(b), z_\alpha(b), d), \qquad C_\beta(b) = \mathcal{R}_\beta(b,d)H_6(c(b), z_\beta(b), d).$$

At $(b_0, z_\alpha(b_0))$, the level-eight shifted block has one transverse pole, labelled by μ, and no other Kac divisor through that point. The analogous statement holds for the level-six block and ν. Therefore the definitions of Q_8, Q_6 give

$$\begin{aligned}H_8(c(b), z_\alpha(b), d) &= \frac{\mathcal{R}_\mu(b,d)}{\delta_\alpha(b)} + Q_8(d) + O(t),\\ H_6(c(b), z_\beta(b), d) &= \frac{\mathcal{R}_\nu(b,d)}{\delta_\beta(b)} + Q_6(d) + O(t).\end{aligned}$$

For either nested pair (κ,τ), expansion of $\mathcal{R}_\kappa\mathcal{R}_\tau/\delta_\kappa$ gives finite term

$$T_{\kappa,\tau} = \frac{\dot{\mathcal{R}}_\kappa\mathcal{R}_\tau + \mathcal{R}_\kappa\dot{\mathcal{R}}_\tau}{\dot\delta_\kappa} - \frac{\mathcal{R}_\kappa\mathcal{R}_\tau\ddot\delta_\kappa}{2\dot\delta_\kappa^2},$$

where every undotted factor on the right is evaluated at b_0. The two t^{-1} terms cancel because

$$\frac{\mathcal{R}_\alpha\mathcal{R}_\mu}{\dot\delta_\alpha} + \frac{\mathcal{R}_\beta\mathcal{R}_\nu}{\dot\delta_\beta} = 0;$$

this is Lemma 7.2 together with $P_2P_8 = P_4P_6$.

Write $y = \Delta - h_0$. Expanding

$$\frac{C_\alpha(b)}{\Delta - h_\alpha(b)} + \frac{C_\beta(b)}{\Delta - h_\beta(b)}$$

at $t = 0$, the cancelled t^{-1} numerator produces the y^{-2} coefficient

$$\frac{\dot h_\alpha - \dot h_\beta}{\dot\delta_\alpha}\mathcal{R}_\alpha(b_0,d)\mathcal{R}_\mu(b_0,d) = -\frac{192}{1121229484375}P_2(d)P_8^{(33/16)}(d).$$

The y^{-1} coefficient is

$$\mathcal{R}_\alpha(b_0, d)Q_8(d) + \mathcal{R}_\beta(b_0, d)Q_6(d) + T_{\alpha,\mu} + T_{\beta,\nu}.$$

The first two residue constants are $-4P_2/7$ and $648P_4/13475$. Differentiating the finite products in Definition 6.1 and factoring over $\mathbb{Q}[d]$ gives

$$T_{\alpha,\mu} + T_{\beta,\nu} = \Theta_{10}(d).$$

Substitution proves the stated principal part. $\square$

Remark 7.5 (Exact replay and its scope)**.** The PBW commutation, Ward recursion, determinant, rank, and null-vector checks for the spin collision in Propositions 5.9 and 5.10, as well as every residue identity in Theorem 5.11, are replayed by `tools/audit_low_levels.py`. The computation uses exact rational arithmetic. The singular vectors, fusion polynomials, 15×15 Gram matrix, Smith-layer dimensions, and intrinsic second crossing scalar for the earlier energy-module grade-three and grade-seven resonances are replayed by `tools/audit_first_resonances.py`. The same script enumerates every repeated Kac class below grade seven and verifies the child-module determinant gaps used in Proposition 6.6. The Kac slopes, four generic residue constants, fusion factorization, generic-recursion evaluation of γ_{10}, and polynomial Θ_{10} are replayed by `tools/audit_grade10_confluence.py`. The shifted Laurent recursion and its first two collision moments are independently replayed by `tools/audit_confluent_recursion.py`. The grade-ten confluence script checks the algebraic confluence calculation; it does not construct the 42×42 Gram matrix, χ_{10}, or the second crossing form independently. The Smith profile at grade ten follows from the determinant and embedding-diagram argument in Proposition 7.1.

Two further scripts test the principal-part theorems directly against the inverse-Shapovalov definition rather than against the confluence algebra. `tools/audit_direct_principal_parts.py` reconstructs H_N, $N \leq 7$, exactly as rational functions of Δ at sample external weights and verifies Theorems 5.8 and 5.11, the complete grade-seven principal part of Theorem 6.7 including the explicit Θ_7, the pole-order claims of Proposition 6.6, the table of Section 9, and the closed forms of Q_4^E, Q_5^E. `tools/audit_grade10_direct.py` performs the analogous direct 42×42 computation at grade ten: it verifies the kernel dimension and determinant valuation of Proposition 7.1 and the complete principal part of Theorem 7.4, including the value of γ_{10} and the closed forms of Q_8, Q_6. Finally, `tools/audit_stocco_comparison.py` verifies the imported identities of Remark 6.2 at a generic sample point and the agreement with the pole-free expressions of [10] discussed in Section 9.

8 All-order confluent recursion

Definition 8.1 (Shifted Laurent recursion)**.** Let $t = b - b_0$, and work in

$$K((t)), \qquad K = \mathbb{Q}(i\sqrt{3}, d),$$

where d is transcendental: K is the rational-function field in d over the number field $\mathbb{Q}(i\sqrt{3})$. For a positive Kac label $a = (r, s)$, write

$$\ell_a = rs, \qquad h_a(t) = h_{r,s}(b_0 + t), \qquad R_a(t) = \mathcal{R}_{r,s}(b_0 + t, d), \qquad z_a(t) = h_a(t) + \ell_a.$$

Define shifted Laurent coefficients recursively by

$$S_{a,0}(t) = 1$$

and, for $M \geq 1$,

$$S_{a,M}(t) = \sum_{\substack{e=(u,v)\in\mathbb{Z}^2_{>0} \\ \ell_e=uv\leq M}} \frac{R_e(t)}{z_a(t) - h_e(t)} S_{e,M-\ell_e}(t).$$

Every denominator in this formula is a nonzero rational function of b by Lemma 6.3, so each summand has a well-defined Laurent expansion, and the recursion terminates because its second index strictly decreases.

Lemma 8.2 (A priori shifted valuation bound)**.** *For a positive label e, and for two positive labels a, e, set*

$$\rho_e = \max\{0, -\operatorname{ord}_t R_e(t)\}, \qquad \sigma_{a,e} = \max\{0, \operatorname{ord}_t(z_a(t) - h_e(t))\}.$$

Define nonnegative integers $W_{a,M}$ recursively by

$$W_{a,0} = 0,$$

and, for $M \geq 1$,

$$W_{a,M} = \max\left(\{0\} \cup \{\rho_e + \sigma_{a,e} + W_{e,M-\ell_e} : \ell_e \leq M\}\right).$$

Then

$$\operatorname{ord}_t S_{a,M}(t) \geq -W_{a,M}.$$

In particular, $W_{a,M}$ is an a priori pole bound computed solely from the finite recursion tree and the valuations of its explicit residue and denominator factors.

Proof. The assertion is clear for $M = 0$. If it holds at lower grades, the e-summand in Definition 8.1 has valuation at least

$$-\rho_e - \sigma_{a,e} - W_{e,M-\ell_e}.$$

The valuation of a finite sum is at least the minimum of the summand valuations. Taking the displayed maximum proves the induction step. All sets involved are finite because $\ell_e \leq M$. $\square$

Definition 8.3 (Collision moments)**.** For $N \geq 1$, set

$$\mathcal{C}_{n,N} = \{a = (r, s) \in \mathbb{Z}^2_{>0} : \ell_a \leq N,\ |4r - 3s| = n\}.$$

For $a \in \mathcal{C}_{n,N}$, define

$$B_{a,N}(t) = R_a(t) S_{a,N-\ell_a}(t), \qquad \delta_a(t) = h_a(t) - \lambda_n.$$

If $\delta_a \not\equiv 0$ and $B_{a,N} \not\equiv 0$, put

$$\varepsilon_a = \operatorname{ord}_t \delta_a, \qquad v_{a,N} = \max\{0, -\operatorname{ord}_t B_{a,N}\}.$$

If $\delta_a \not\equiv 0$ and $B_{a,N} \equiv 0$, put $\varepsilon_a = \operatorname{ord}_t \delta_a$ and $v_{a,N} = 0$. If $\delta_a \equiv 0$, set $\varepsilon_a = \infty$ and $v_{a,N} = 0$. Define also the a priori numerator bound

$$\widehat{v}_{a,N} = \rho_a + W_{a,N-\ell_a}.$$

Lemma 8.2 gives

$$v_{a,N} \leq \widehat{v}_{a,N}.$$

Define the exact finite structural pole bound

$$M_{n,N} = \max_{a\in\mathcal{C}_{n,N}} \begin{cases} 1 + \lfloor v_{a,N}/\varepsilon_a \rfloor, & \varepsilon_a < \infty, \\ 1, & \varepsilon_a = \infty, \end{cases}$$

and its a priori majorant

$$\widehat{M}_{n,N} = \max_{a \in \mathcal{C}_{n,N}} \begin{cases} 1 + \lfloor \widehat{v}_{a,N}/\varepsilon_a \rfloor, & \varepsilon_a < \infty, \\ 1, & \varepsilon_a = \infty. \end{cases}$$

Thus $M_{n,N} \le \widehat{M}_{n,N}$. The former can be sharpened after the exact rational germs have been computed, while the latter bounds the number of moments before those cancellations are simplified. For $1 \le k \le M_{n,N}$, define the *collision moments*

$$\widetilde{A}_{n,k,N}(d) = \mathrm{CT}_t \sum_{a \in \mathcal{C}_{n,N}} B_{a,N}(t)\delta_a(t)^{k-1}.$$

Here CT_t denotes the coefficient of t^0 in a Laurent series. Empty collision classes contribute zero. Put

$$m_{n,N} = \max\bigl(\{k : \widetilde{A}_{n,k,N} \neq 0\} \cup \{0\}\bigr).$$

It satisfies $m_{n,N} \le M_{n,N}$, and Theorem 8.6 identifies it with the exact multiplicity of the pole of H_N at λ_n.

Lemma 8.4 (Backward Laurent-precision budget)**.** *Put*

$$D_{a,e}(t) = \frac{R_e(t)}{z_a(t) - h_e(t)}, \qquad p_{a,e} = \rho_e + \sigma_{a,e}.$$

Suppose $S_{a,M}$ is required through the coefficient of t^U. For its e-summand it is sufficient to know

$$D_{a,e} \quad \textit{through } t^{U+W_{e,M-\ell_e}}, \qquad S_{e,M-\ell_e} \quad \textit{through } t^{U+p_{a,e}}. \tag{8.1}$$

Iterating these requests down the strictly grade-decreasing recursion tree produces a finite a priori upper precision at every node.

In particular, to compute every collision moment at grade N, it is sufficient for each a to compute R_a through $t^{W_{a,N-\ell_a}}$, $S_{a,N-\ell_a}$ through t^{ρ_a}, and δ_a^{k-1} through $t^{\widehat{v}_{a,N}}$. Thus the valuation majorant together with (8.1) *is an actual finite jet budget, not only a pole-order bound.*

Proof. The lower bounds are

$$\mathrm{ord}_t\, D_{a,e} \ge -p_{a,e}, \qquad \mathrm{ord}_t\, S_{e,M-\ell_e} \ge -W_{e,M-\ell_e}.$$

In a product coefficient of degree at most U, the first bound means that no coefficient of the second factor above degree $U + p_{a,e}$ can occur; the second means that no coefficient of the first factor above degree $U + W_{e,M-\ell_e}$ can occur. This proves (8.1). The recursion tree is finite because the remaining grade decreases along every edge.

For $B_{a,N} = R_a S_{a,N-\ell_a}$ through degree zero, the same product argument gives the stated precisions $W_{a,N-\ell_a}$ and ρ_a. Its lower degree is at least $-\widehat{v}_{a,N}$, so only coefficients of δ_a^{k-1} through degree $\widehat{v}_{a,N}$ can enter a constant term. □

Lemma 8.5 (Separated-pole Laurent cancellation)**.** *Let $\mu_1, \ldots, \mu_s$ be distinct, and let*

$$f_j(t, \Delta) \in K(\Delta)((t)),$$

that is, a Laurent series in t whose coefficients are rational functions of Δ. Suppose that every t-coefficient of f_j is a proper rational function of Δ whose finite poles, if any, lie only at $\Delta = \mu_j$. If $\sum_{j=1}^{s} f_j(t, \Delta)$ has no negative power of t, then neither does any $f_j(t, \Delta)$.

Proof. For each $m < 0$, the coefficient of t^m in the sum is zero. It is a sum of proper rational functions with pairwise disjoint pole sets. Uniqueness of partial fractions forces every summand to vanish. Repeating this for the finitely many possible negative exponents proves the claim. □

Theorem 8.6 (Pole-free Ising recursion and all-order equality). *For every $N \geq 1$,*

$$H_N(1/2, \Delta, d) = \sum_{\substack{n \geq 0 \\ \mathcal{C}_{n,N} \neq \varnothing}} \sum_{k=1}^{M_{n,N}} \frac{\widetilde{A}_{n,k,N}(d)}{(\Delta - \lambda_n)^k}.$$

The sums and every constant-term calculation are finite, $\widetilde{A}_{n,k,N}(d) \in \mathbb{Q}[d]$, and the recursion in Definitions 8.1–8.3 terminates after grades strictly below N. In particular, by the uniqueness in Theorem 4.2, the collision moments are the partial-fraction coefficients of the block,

$$\widetilde{A}_{n,k,N} = A_{n,k,N},$$

and $m_{n,N}$ is the exact multiplicity of the pole of H_N at λ_n. Moreover

$$m_{n,N} \leq M_{n,N} \leq \widehat{M}_{n,N},$$

where $\widehat{M}_{n,N}$ is computed before simplifying cancellations inside the shifted blocks. The backward rule of Lemma 8.4 supplies a finite truncation order for every intermediate Laurent germ. Consequently, with $A_{n,k}(d;q)$ as in Theorem 4.2, one has the coefficientwise locally finite identity

$$H_\Delta^{(1/2)}(d \mid q) = 1 + \sum_{n \geq 0} \sum_{k \geq 1} \frac{A_{n,k}(d;q)}{(\Delta - \lambda_n)^k}.$$

This series agrees coefficient by coefficient with Definition 2.2, hence with the inverse-Shapovalov definition of the torus one-point block.

Proof. For generic b, Proposition 6.4 gives

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{R_a(t)}{\Delta - h_a(t)} H_{N-\ell_a}(c(b), z_a(t), d).$$

Induction on M in the same recursion gives

$$S_{a,M}(t) = H_M(c(b), z_a(t), d)$$

for generic b. Thus

$$H_N(c(b), \Delta, d) = \sum_{\ell_a \leq N} \frac{B_{a,N}(t)}{\Delta - h_a(t)}. \tag{8.2}$$

All sums are finite.

Fix N, and choose Δ outside the finite set of Ising Kac weights occurring at grade N. By Lemma 2.4, the inverse-Shapovalov expression is analytic in t near zero: its specialized Gram determinant is nonzero at this Δ. Equation (8.2), initially valid for generic b, is an identity of rational functions and therefore holds on the punctured formal neighborhood of b_0. This applies for every Δ in the complement of the finite exceptional set; hence each negative t-coefficient vanishes as a rational function of Δ, not merely after one numerical specialization.

Group its right-hand side according to the distinct limits λ_n:

$$\mathcal{G}_{n,N}(t, \Delta) = \sum_{a \in \mathcal{C}_{n,N}} \frac{B_{a,N}(t)}{\Delta - h_a(t)}.$$

Every t-coefficient of $\mathcal{G}_{n,N}$ is a proper rational function of Δ with its only possible finite pole at $\Delta = \lambda_n$. The sum over n is analytic in t, as just proved. Lemma 8.5 therefore shows that each $\mathcal{G}_{n,N}$ is separately regular in t. Hence every collision class has a finite algebraic confluent value.

Put $x = \Delta - \lambda_n$. In the formal expansion at $x = \infty$,

$$\frac{1}{\Delta - h_a(t)} = \frac{1}{x - \delta_a(t)} = \sum_{j \geq 0} \frac{\delta_a(t)^j}{x^{j+1}}.$$

Taking CT_t after summing the collision class shows that the coefficient of x^{-k} is exactly

$$\mathrm{CT}_t \sum_{a \in \mathcal{C}_{n,N}} B_{a,N}(t)\delta_a(t)^{k-1} = \widetilde{A}_{n,k,N}(d).$$

If $\delta_a \not\equiv 0$, then the summand has positive t-valuation whenever $k > 1 + \lfloor v_{a,N}/\varepsilon_a \rfloor$, so its constant term vanishes. If $\delta_a \equiv 0$, it vanishes identically for $k > 1$. This proves the bound $M_{n,N}$, the finiteness of the pole sum, and the finite-jet nature of every constant-term calculation. Independently of cancellations, Lemma 8.2 replaces $v_{a,N}$ by the a priori bound $\widehat{v}_{a,N}$, proving $M_{n,N} \leq \widehat{M}_{n,N}$. Lemma 8.4 then propagates the required upper precision backward through the finite recursion tree, fixing a finite jet budget before the exact germs are simplified. In particular, since $\mathrm{ord}_t\, \delta_a = \varepsilon_a > 0$, only finitely many terms of the formal x^{-1}-expansion can contribute to CT_t; this justifies interchanging the constant term with that expansion. The confluent value of each class tends to zero as $\Delta \to \infty$ and has no finite pole except λ_n; therefore these finitely many moments are its complete partial-fraction expansion, with no polynomial remainder.

Every $R_a(t)$ is polynomial in d with coefficients rational in t, and all shifted denominators are independent of d. Therefore every collision moment is polynomial in d. Equation (8.2) and the analyticity argument identify the resulting rational function with the specialized inverse-Shapovalov coefficient; by the uniqueness in Theorem 4.2, this gives $\widetilde{A}_{n,k,N} = A_{n,k,N}$, and $m_{n,N}$ is the exact pole multiplicity of H_N at λ_n. The specialized coefficient has rational coefficients at $c = 1/2$; uniqueness of its partial fractions improves $\widetilde{A}_{n,k,N} \in K[d]$ to $\widetilde{A}_{n,k,N} \in \mathbb{Q}[d]$. Finally, local finiteness in q is Corollary 3.3, completing the all-order identity. □

Corollary 8.7 (Recovery of the explicit confluent formulas). *The $N = 3$ and $N = 7$ moments in the $\lambda_5 = 1/2$ class give Theorems 5.8 and 6.7. The $N = 4$ and $N = 10$ moments in the $\lambda_2 = 1/16$ class give Theorems 5.11 and 7.4.*

Proof. At $N = 3$, the class $\mathcal{C}_{5,3}$ consists of $(2,1)$ and $(1,3)$. Its first moment is

$$\frac{9}{28} E_2(d) H_1(1/2, 5/2, d) - \frac{8}{105} E_3(d),$$

and its higher moments vanish. At $N = 7$, the same class has the two nested shifted poles $(2,1) \to (1,5)$ and $(1,3) \to (4,1)$. Its first two moments are the simple- and double-pole coefficients of Theorem 6.7.

At $N = 4$, the class $\mathcal{C}_{2,4}$ consists of $(1,2)$ and $(2,2)$, and both B-terms are regular at $t = 0$. Its first moment is

$$-\frac{4}{7} P_2(d) H_2(1/2, 33/16, d) + \frac{648}{13475} P_4(d),$$

while all higher moments vanish.

At $N = 10$, the same spin collision class has

$$B_{(1,2),10} = \mathcal{R}_{1,2} H_8(c(b), h_{1,2} + 2, d), \qquad B_{(2,2),10} = \mathcal{R}_{2,2} H_6(c(b), h_{2,2} + 4, d).$$

Their zeroth and first collision moments are exactly the two Laurent coefficients computed in the proof of Theorem 7.4; all higher moments vanish by the grade-ten Smith profile. □

9 Low-grade output and comparison with earlier expansions

Through grade four, every pole of the Ising block is simple: $m_{n,N} = 1$ for all classes with $N \le 4$, by Proposition 6.6. The complete list of nonzero collision moments through grade four, computed from Definition 8.3, equal to the partial-fraction coefficients $A_{n,k,N}$ by Theorem 8.6, and factored over $\mathbb{Q}[d]$, is

$$
\begin{aligned}
A_{1,1,1} &= \tfrac{1}{2}\, d(d-1),\\
A_{1,1,2} &= \tfrac{1}{4}\, d^2(d-1)^2,\\
A_{2,1,2} &= -\tfrac{1}{7}\, d(d-1)(2d-7)(2d-1),\\
A_{5,1,2} &= \tfrac{1}{28}\, d(d-1)(3d-14)(3d-5),\\
A_{1,1,3} &= \tfrac{1}{120}\, d^2(d-1)^2(17d^2-113d+236),\\
A_{2,1,3} &= -\tfrac{8}{231}\, d^2(d-1)^2(2d-7)(2d-1),\\
A_{5,1,3} &= -\tfrac{1}{84}\, d(d-1)(d^4-50d^3+311d^2-550d+210),\\
A_{9,1,3} &= \tfrac{3}{3080}\, d(d-11)(d-6)(d-1)(3d-14)(3d-5),\\
A_{1,1,4} &= -\tfrac{1}{1440}\, d^2(d-1)^2(31d^4-830d^3+5567d^2-13408d+8460),\\
A_{2,1,4} &= \tfrac{1}{40425}\, d(d-1)(2d-7)(2d-1)\\
&\qquad \times (248d^4-5752d^3+25274d^2-33630d+5775),\\
A_{5,1,4} &= -\tfrac{1}{61152}\, d^2(d-1)^2(149d^4-2506d^3-2679d^2+70508d-106540),\\
A_{8,1,4} &= -\tfrac{2}{20475}\, d(d-13)(d-6)(d-1)(2d-15)(2d-7)(2d-5)(2d-1),\\
A_{9,1,4} &= \tfrac{9}{86240}\, d^2(d-11)(d-6)(d-1)^2(3d-14)(3d-5),\\
A_{13,1,4} &= \tfrac{1}{184800}\, d(d-20)(d-13)(d-11)(d-6)(d-1)(3d-14)(3d-5).
\end{aligned}
$$

The entries are consistent with the local theorems: $A_{5,1,3}$ is the residue of Theorem 5.8, $A_{2,1,4}$ is the polynomial of Theorem 5.11, and $A_{13,1,4} = \mathcal{A}_{4,1}(b_0)\mathcal{P}_{4,1}(b_0, d)$. All entries are verified against the inverse-Shapovalov definition by `tools/audit_direct_principal_parts.py`.

Remark 9.1 (Comparison with the pole-free expressions of Stocco)**.** Stocco [10, Sections II.2–II.3 and Appendix A] writes the generic torus recursion over a common denominator,

$$
H_\Delta^{(N)} = \frac{R_{1,1}\, K_N(\Delta, \Delta_1, \beta)}{\prod_{kl\le N}(\Delta - \Delta_{\langle k,l\rangle})},
$$

and gives closed pole-free numerators K_N for $N \le 4$: order two in equation (II.11), orders three and four in Appendix A. In our notation $\beta = b^2$, $\Delta_1 = d$, and his $\Delta_{\langle r,s\rangle}$ is $h_{r,s}(b)$. We have verified in exact arithmetic that his order-two and order-three numerators reproduce the generic blocks H_2 and H_3 of Definition 2.2 identically in b, and that his order-four numerator reproduces H_4 at sample values of b and at $b = b_0$; see `tools/audit_stocco_comparison.py`. (The identical-in-b comparison was run only through order three because of the cost of expanding the order-four rational functions; the sampled order-four check includes the specialization $b = b_0$ used in this paper.) In particular, specializing his expressions at $\beta = b_0^2 = -4/3$ reproduces the $N \le 4$ table above. At $c = 1/2$, collided labels make his denominator vanish to higher order (for instance $\Delta_{\langle 2,1\rangle} = \Delta_{\langle 1,3\rangle} = \frac{1}{2}$ at order three), and the specialized numerators $K_N(\Delta, d, -4/3)$ acquire the compensating zeros, so that the actual pole multiplicities are the $m_{n,N}$ of Definition 8.3. The present paper upgrades these closed low-order expressions to an all-order fixed-c statement with a termination and equality proof.

10 Irreducible Ising specialization

Proposition 10.1 (Identity insertion in the Verma recursion)**.** *For $d = 0$,*

$$H^{(1/2)}_{\Delta}(0 \mid q) = 1$$

as a rational function of Δ. Thus specialization of the meromorphic Verma block at a degenerate weight is not the irreducible Ising trace; the latter requires taking the quotient by the radical.

Proof. In Definition 6.1, the factor with $k = l = 1$ in $\mathcal{P}_{r,s}(b, 0)$ is

$$\frac{(b + b^{-1})^2 - (b + b^{-1})^2}{4} = 0.$$

Hence every generic residue $R_a(t)$ vanishes identically at $d = 0$. Definitions 8.1 and 8.3 then give $\widetilde{A}_{n,k,N}(0) = 0$, hence $A_{n,k,N}(0) = 0$, for every $N \geq 1$. Equivalently, in the inverse-Shapovalov definition the identity insertion has $\rho_N = G_N$, so $F_N = p(N)$ and eta reduction gives $H = 1$. □

Definition 10.2 (Eta-reduced irreducible identity block)**.** For $h \in \{0, 1/2, 1/16\}$, let $L(1/2, h)$ be the irreducible quotient of $M(1/2, h)$, and define

$$\widehat{H}_h(q) = \prod_{m\geq 1}(1 - q^m) \sum_{N\geq 0} \dim L(1/2, h)_{h+N}\, q^N.$$

Theorem 10.3 (The three Ising characters)**.** *The quotient identity blocks are*

$$\widehat{H}_0(q) = \sum_{j\in\mathbb{Z}} \left(q^{((24j+1)^2-1)/48} - q^{((24j+7)^2-1)/48}\right),$$

$$\widehat{H}_{1/2}(q) = \sum_{j\in\mathbb{Z}} \left(q^{((24j+5)^2-25)/48} - q^{((24j+11)^2-25)/48}\right),$$

and

$$\widehat{H}_{1/16}(q) = \sum_{j\in\mathbb{Z}} \left(q^{((24j-2)^2-4)/48} - q^{((24j+10)^2-4)/48}\right).$$

Consequently,

$$\chi_h(q) = q^{h-1/48} \frac{\widehat{H}_h(q)}{\prod_{m\geq 1}(1 - q^m)}$$

are the vacuum, energy, and spin Ising characters. Their eta-reduced beginnings are

$$\begin{aligned}
\widehat{H}_0 &= 1 - q - q^6 + q^{11} + q^{13} - q^{20} + \cdots, \\
\widehat{H}_{1/2} &= 1 - q^2 - q^3 + q^7 + q^{17} - q^{25} - q^{28} + \cdots, \\
\widehat{H}_{1/16} &= 1 - q^2 - q^4 + q^{10} + q^{14} - q^{24} - q^{30} + \cdots.
\end{aligned}$$

Proof. The Ising model is the minimal series $c_{3,4} = 1/2$. The Class R^+, Case 1+, BGG resolution in [7, Theorem 6.9] is an exact complex of Verma modules. Taking its Euler–Poincaré character gives, for an interior label (r, s),

$$\chi_{r,s}(q) = \frac{1}{\eta(q)} \sum_{j\in\mathbb{Z}} \left(q^{(24j+4r-3s)^2/48} - q^{(24j+4r+3s)^2/48}\right).$$

Use $(r, s) = (1, 1), (2, 1), (1, 2)$, respectively. Since

$$h_{r,s} = \frac{(4r - 3s)^2 - 1}{48}, \qquad \eta(q) = q^{1/24} \prod_{m\geq 1}(1 - q^m),$$

factoring out $q^{h_{r,s}-1/48}$ gives exactly the three displayed $\widehat{H}$-series. For any fixed power of q, only finitely many integers j contribute, so the Euler–Poincaré sums are coefficientwise finite. Direct evaluation of the exponents through degree 30 gives the final three lines. □

Remark 10.4 (Character replay)**.** The three BGG numerators and the resulting quotient dimensions through grade 30 are replayed by `tools/audit_ising_characters.py` using exact integer arithmetic, and are checked there against an independent free-fermion count: the spin module is counted by partitions into distinct positive integers, and the vacuum and energy modules by partitions of $2N$ (respectively $2N+1$) into distinct odd parts with an even (respectively odd) number of parts.

11 Scope and outlook

The all-order result is a coefficientwise theorem for the meromorphic Verma-module torus block at $c = 1/2$. Its output is fixed at the Ising central charge, while its construction uses the exact formal deformation $b = b_0 + t$ to combine resonant generic-recursion terms. The method should therefore be distinguished both from numerical regularization and from a hypothetical recursion formulated solely in intrinsic data of the specialized category.

At any prescribed grade the label sets and recursion trees are finite. Lemmas 8.2 and 8.4 give an a priori Laurent jet budget, but do not claim a favorable complexity bound: exact rational simplification can grow rapidly with the grade. The local Smith analysis explains the first simple and double poles, but an all-depth identification of every Smith layer with the Ising embedding diagram is not required for the coefficientwise confluence theorem. The grade-ten principal part has been verified directly against the inverse-Shapovalov definition (Remark 7.5); an independent symbolic construction of the grade-ten 42×42 second crossing form itself would nevertheless be a useful additional audit.

For a general minimal-model charge $c_{p,p'}$, the same separated-pole argument applies once the corresponding collision classes and residue valuations are inserted. Making that extension uniform in p, p', and finding an intrinsic fixed-central-charge recursion that eliminates the formal deformation altogether, remain open directions.

Code availability

All exact-arithmetic audit scripts cited in the text are distributed with this paper as ancillary files: the eight `tools/audit_*.py` scripts described in Remarks 6.2 and 7.5 and in the character-replay remark, together with the shared library modules `tools/exact_shapovalov.py` and `tools/direct_reconstruction.py` on which they depend. All computations use exact rational arithmetic (Python 3 with SymPy; the pinned dependency is listed in `requirements-audit.txt`). Running `make audit` replays every check except the direct grade-ten reconstruction, which the slower target `make audit-deep` adds.

A The level-four Gram matrix at $c = 1/2$

In the ordered PBW basis $(L_{-4},\, L_{-3}L_{-1},\, L_{-2}^2,\, L_{-2}L_{-1}^2,\, L_{-1}^4)|\Delta\rangle$ of Proposition 5.9, direct Virasoro commutation gives

$$G_4 = \begin{pmatrix} 8\Delta + \frac{5}{2} & 14\Delta & 24\Delta + \frac{3}{2} & 36\Delta & 120\Delta \\ 14\Delta & 12\Delta^2 + 14\Delta & 30\Delta & 40\Delta^2 + 20\Delta & 192\Delta^2 + 48\Delta \\ 24\Delta + \frac{3}{2} & 30\Delta & 32\Delta^2 + 36\Delta + \frac{17}{8} & 48\Delta^2 + 51\Delta & 216\Delta^2 + 144\Delta \\ 36\Delta & 40\Delta^2 + 20\Delta & 48\Delta^2 + 51\Delta & 32\Delta^3 + 118\Delta^2 + 33\Delta & 288\Delta^3 + 336\Delta^2 + 96\Delta \\ 120\Delta & 192\Delta^2 + 48\Delta & 216\Delta^2 + 144\Delta & 288\Delta^3 + 336\Delta^2 + 96\Delta & 384\Delta^4 + 1152\Delta^3 + 1056\Delta^2 + 288\Delta \end{pmatrix}$$

at $c = 1/2$.

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